Quantifying metacognition with the `hmetad` package in R

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Abstract

A longstanding goal in metacognition research is the quantification of confidence-related parameters independently from task performance. While there is now a proliferation of generative models of confidence in numerous psychological tasks, the meta-d' model remains a popular approach to estimating metacognitive sensitivity and metacognitive bias within a signal detection theoretic framework. However, as questions in metacognition research increase in complexity, it remains an ongoing challenge to develop extensions of the meta-d' model built to deal with longitudinal, repeated-measures, or other complex experimental designs. Here we introduce the open-source R package `hmetad`, which allows users to fit the meta-d' model for a wide range of experimental designs using a familiar formula syntax. We also review how the meta-d' model can be validated in a Bayesian workflow which investigates posterior estimates of key parameters. Overall, the `hmetad` package offers a comprehensive interface to fitting, evaluating, and visualizing signal detection theoretic models of metacognition.

Keywords: metacognition, Bayesian statistics, multilevel modeling

Metacognition—the capacity to monitor and control one's cognitive processes—is increasingly recognized as a core component of how we form and revise beliefs, how we make flexible decisions on the basis of incomplete evidence, how we dynamically gather information, and how we interact with others (Fleming, 2024; Peters, 2024; Rahnev, 2021). Progress in metacognition research is contingent upon the development of psychometric tools which isolate metacognitive contributions to behavior from performance-related factors. As a simple example, consider two participants (A and B) in a memory study, with participant A exhibiting a stronger correlation between memory accuracy and confidence. At first glance, this might appear to suggest that participant A has more sensitive metacognition. But if it is also found that participant A has better memory overall, then any observed difference in metacognition may simply be due to this difference in task performance (Fleming & Lau, 2014).

In response to this challenge of isolating metacognitive capacity from other aspects of task performance, researchers studying metacognition have developed a range of models and measures that explicitly aim to control for performance confounds (Rahnev, 2025). Measures typically target either metacognitive efficiency (i.e., the strength of the relationship between confidence ratings and accuracy relative to task performance) or metacognitive bias (i.e., the overall level of confidence), which are themselves potential confounds for each other. One such model—the meta-d' model—builds on Signal Detection Theory to estimate two sensitivity parameters. In this model, d' reflects participants' sensitivity to the stimulus when making a first-order response to the task, whereas meta-d' reflects the same sensitivity expressed through confidence ratings (Maniscalco & Lau, 2012, 2014). Comparing the two sensitivities results in a measure of metacognitive efficiency, $\frac{\text{meta-}d'}{d'}$, commonly known as M-ratio (Fleming & Lau, 2014). The meta-d' model is currently the most widely used measure of metacognition, partly due to the implementation of hierarchical Bayesian estimation routines that provide stronger statistical power and less estimation bias than single-participant model fits

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(Fleming, 2017).

Despite these advances, there remain significant barriers to modeling metacognition. Recent evidence indicates that the separation of metacognitive factors assumed by the meta-d' model is not guaranteed (e.g., Guggenmos, 2021; Rausch et al., 2026; Xue et al., 2021). Moreover, existing implementations of the model are specific to simple experimental designs, meaning that users must develop their own analysis software to accommodate more complex designs. Previous implementations also have limited convergence diagnostics, making it sometimes unclear whether the approximations used by fitted models are sufficient for well-calibrated inference, and lack direct interfaces with other statistical software, complicating the computation of rigorous statistical tests using model outputs. Although it has been suggested that the study of metacognition should benefit from models including trial-level predictors (e.g., neural activity or previous confidence level), trial-level models have been too cumbersome to fit in practice (Fleming, 2017). Finally, although significant attention has been paid to measures of metacognitive efficiency, less research has focused on metacognitive bias, let alone modeling both efficiency and bias in a single comprehensive framework.

Here we introduce the `hmetad` package, a new open-source platform for hierarchical Bayesian estimation of the meta-d' model in the programming language R. As an extension of the more general statistical package `brms`, the `hmetad` package provides a simple formula syntax to specify arbitrary model designs, thorough model convergence diagnostics, and interfacing with other packages for model validation, statistical testing, and visualization in R. Beyond several improvements to computational efficiency, the `hmetad` package also computes model estimates and empirical values of a range of imputed quantities popular in metacognition research, including various response probabilities and Receiver Operating Characteristic (ROC) curves. Moreover, the `hmetad` package jointly models task performance, response bias, metacognitive efficiency, as well as meta-Δ, a new performance-corrected measure of metacognitive bias (O'Neill et al., 2026), all as part of a single model.

After outlining the meta-d' model, we demonstrate the functionality of the `hmetad` package by re-analyzing an open dataset investigating associations between mental health and metacognition (Rouault et al., 2018). Rather than recommending a fixed set of defaults, we highlight how a principled Bayesian workflow recommended elsewhere in Bayesian statistics can guide modeling choices (Gelman et al., 2026; Schad et al., 2021). Overall, we argue that the `hmetad` package can help to address concerns in metacognition research about statistical power, reliability, and sensitivity to distributional assumptions, while at the same time opening the door to rigorously addressing a broader set of research questions.

The meta-d' model

In this section, we review the meta-d' model, focusing on its generative mechanism for confidence ratings and the unique parameterization of the model used in the `hmetad` package.

Data

The meta-d' model can be used to describe data from binary choice tasks with ordinal-valued confidence ratings (Figure 1A). That is, given a stimulus $S \in \{0, 1\}$, the participant is asked to make a response $R \in \{0, 1\}$ along with a confidence rating $C \in \{1, \ldots, K\}$ (where K is the maximum confidence rating), such that the participant's decision is correct if $R = S$. For instance, participants might be asked to judge whether a stimulus is old (i.e., presented in a prior study phase) or new, determine whether a random dot motion stimulus is moving left or right, or decide which of two food items contains more calories. Along with the primary decision, participants are asked to rate their confidence in their decision on an ordinal scale indicating whether their decision was a guess, made with high confidence, or somewhere in between.[1] Following Clarke et al. (1959), we refer to the primary decision as the *Type I* task and the task of rating confidence as the *Type II* task.

While it is designed for experimental paradigms with separate Type I and Type II decisions, the meta-d' model is also compatible with paradigms in which the participants make a single joint Type I/Type II response with endpoints representing a confident decision in either of the two Type I responses and with the center representing low confidence ($J \in \{1, \ldots, 2K\}$, see Figure 1B).[2] This design is often advantageous because it takes less time than separate responses and it reduces the likelihood that a participant changes their mind between responses. One can convert

[1] Note that the lower bound on this scale represents guesses, not decisions which the participant is confident are incorrect. Although such changes of mind are of interest to metacognition research, they are incompatible with the meta-d' model and tend to be infrequent in controlled laboratory tasks.

[2] Because it is designed for tasks in which the Type I task is forced-choice, the meta-d' model can only be used when the joint response has an even number of values.

between the separate and joint responses using the equations below, implemented as the functions `type1_response`, `type2_response`, and `joint_response` in the `hmetad` package:

$$R = \texttt{type1_response}(J, K) = [\![J > K]\!]$$

$$C = \texttt{type2_response}(J, K) = \begin{cases} J - K & \text{if } R = 1 \\ K + 1 - J & \text{if } R = 0 \end{cases}$$

$$J = \texttt{joint_response}(R, C, K) = \begin{cases} K + C & \text{if } R = 1 \\ K + 1 - C & \text{if } R = 0 \end{cases}$$

Figure 1

Two example tasks amenable to analysis with the meta-d′ model. (A) *A signal detection task with a separate Type I and Type II response.* (B) *A two-alternative forced-choice (2AFC) task with a combined Type I/Type II response. In both tasks, the set of responses made on the trial can be decomposed into a binary Type I response and an ordinal Type II response.*



Generative model

Given the design above, the meta-d' model aims to explain the distribution of Type I and Type II responses. Early descriptions of the model stressed that it was not intended to be a generative model; initially, it was simply meant to describe (a) the relation between Type I responses and the stimulus, and (b) the relation between Type II responses and the accuracy of the Type I responses. However, over time this characterization has shifted to an understanding that the meta-d' model is sensitive to a number of generative assumptions, indicating that it is not in fact a theory-free model (Guggenmos, 2021; Rausch et al., 2026). Keeping this in mind, we next overview the statistical process underlying the meta-d' model Figure 2, identifying relevant assumptions along the way.

Type I responses

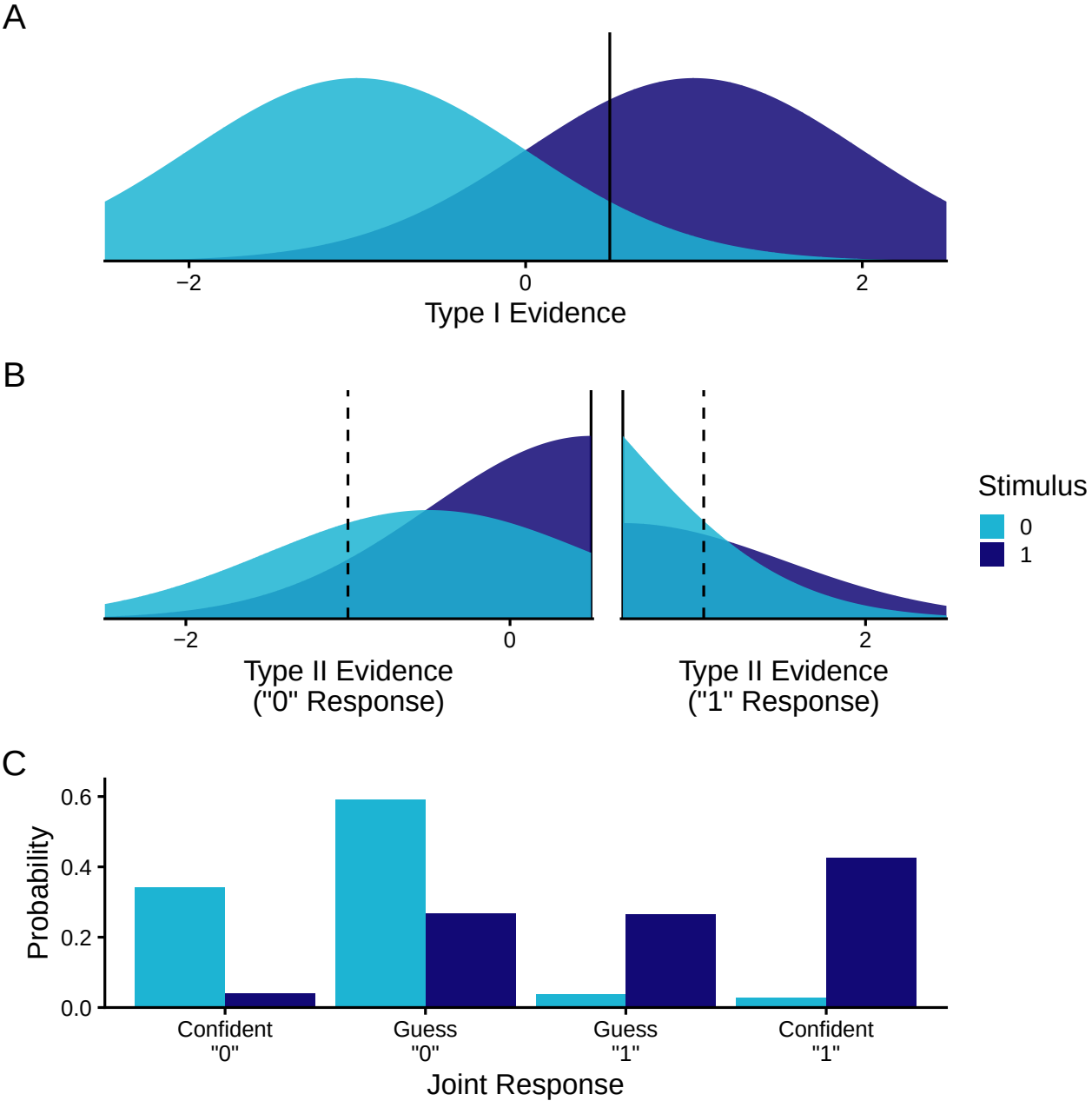
As the meta-d' model is an extension of Signal Detection Theory (SDT), its model for Type I responses is simply the equal variance signal detection model (Clarke et al., 1959; Green & Swets, 1966). Given the stimulus S, equal variance SDT assumes that the observer is presented with a noisy signal $x_1 \sim \mathcal{D}_S(d')$ following a distribution $\mathcal{D}$ parameterized by the observer's sensitivity to the stimulus d' Figure 2A. Although other options are available, it is common to assume that $\mathcal{D}$ is a normal distribution with unit variance, such that:

$$\mathcal{D}_0(d') = \mathcal{N}\left(-\frac{d'}{2}, 1\right)$$

$$\mathcal{D}_1(d') = \mathcal{N}\left(+\frac{d'}{2}, 1\right)$$

Figure 2

Generative process underlying the meta-d′ model. (A) *The observer receives an evidence signal which noisily encodes the stimulus with sensitivity d′. They make a Type I response by comparing the evidence to a response criterion c.* (B) *Depending on the Type I response, the observer receives a second piece of evidence about the stimulus, now with sensitivity meta-d′. They make a confidence rating by comparing the evidence to a set of ordered confidence criteria,* ***meta-*** $\mathbf{c}_2^{\mathbf{R}}$. (C) *Trials are categorized depending on the stimulus and joint Type I/Type II response, where the probability of each trial type is determined by the product of area under the Type I and Type II evidence distribution curves.*



In the Type I task, the observer is meant to determine the underlying value of S on the basis of its noisy signal x_1. To make a response, then, it is assumed that the observer thresholds the signal x_1 by applying a criterion c:

$$R = [\![x_1 > c]\!]$$

Rather than modeling x_1 directly, it is more efficient to marginalize over it to compute the implied probability distribution over Type I responses. Where $F_{\mathcal{D}}$ is the cumulative distribution function of $\mathcal{D}$, this process entails that the observer makes Type I responses with probability:

$$P(R = r \mid S = s) = \begin{cases} 1 - F_{\mathcal{D}_s(d')}(c) & \text{if } r = 1 \\ F_{\mathcal{D}_s(d')}(c) & \text{if } r = 0 \end{cases}$$

For example, the observer's Type I hit rate is $P(R = 1 \mid S = 1) = 1 - F_{\mathcal{D}_1(d')}(c)$ and their Type I false alarm rate is $P(R = 1 \mid S = 0) = 1 - F_{\mathcal{D}_0(d')}(c)$.

Type II responses

In the Type II task, the observer is asked to determine their confidence in their Type I response. A crucial aspect of the meta-d' model is that it does not assume that the observer has access to the same information for the Type I and Type II tasks: it could be the case that the observer gained or lost information about the stimulus after the Type I response was made. As a result, the meta-d' model assumes that the observer makes Type II responses using a separate decision variable (Figure 2B) following a truncated distribution (with superscripts indicating truncation bounds):

$$x_2 \sim \begin{cases} \mathcal{D}_S^{(-\infty, \text{meta-}c]}(\text{meta-}d') & \text{if } R = 0 \\ \mathcal{D}_S^{[\text{meta-}c, \infty)}(\text{meta-}d') & \text{if } R = 1 \end{cases}$$

This Type II decision variable has two important characteristics. First, parameters used to generate Type II responses are named with the prefix "meta-" to distinguish them from parameters used to generate the Type I responses. Second, x_2 is upper-bounded by meta-c when $R = 0$ and lower-bounded by meta-c when $R = 1$ so that x_2 cannot contradict the Type I response implied by x_1 (i.e., the meta-d' model does not allow for changes of mind between Type I and Type II responses). Then, to determine the confidence level $C \in \{1 \ldots K\}$, the observer rates confidence using a set of $K - 1$ ordered confidence criteria, $\mathbf{meta\text{-}c_2^R}$:

$$C = \begin{cases} 1 + \Sigma_{k=1}^{K-1}[x_2 < \text{meta-}c_{2,k}^0] & \text{if } R = 0 \\ 1 + \Sigma_{k=1}^{K-1}[x_2 > \text{meta-}c_{2,k}^1] & \text{if } R = 1 \end{cases}$$

As with the Type I response, it is most efficient to marginalize over x_2 to compute the probability of a Type II response given the stimulus and Type I response:

$$P(C = c \mid S = s, R = r) = \begin{cases} \frac{F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c) - F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,1}^0\right)}{F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 0, c = 1 \\ \frac{F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,k}^0\right) - F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,k+1}^0\right)}{F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 0, 1 \le c \le K \\ \frac{F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,K}^0\right)}{F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 0, c = K \\ \frac{F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,1}^1\right) - F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)}{1 - F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 1, c = 1 \\ \frac{F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,k+1}^1\right) - F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,k}^1\right)}{1 - F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 1, 1 \le c \le K \\ \frac{1 - F_{\mathcal{D}_S(\text{meta-}d')}\left(\text{meta-}c_{2,K}^1\right)}{1 - F_{\mathcal{D}_S(\text{meta-}d')}(\text{meta-}c)} & \text{if } r = 1, c = K \end{cases}$$

In each of these equations, the numerator is the probability that x_2 lies between the thresholds for confidence level c, and the denominator is a normalization term to deal with the fact that x_2 is truncated above or below by meta-c.

Joint likelihood

The meta-d' model aims to explain the joint distribution of Type I and Type II responses, which can conveniently be factored into a product of the Type I and Type II response probabilities (Figure 2C):

$$P(R = r, C = c \mid S = s) = P(R = r \mid S = s)\ P(C = c \mid S = s, R = r)$$

Note that because this distribution is conditional on the stimulus S, the meta-d' model does not model the base rate $P(S = s)$. We consider two specifications of the model likelihood. When trial-level effects are of interest, one can fit the meta-d' model to individual trials using a categorical distribution. Where $\mathcal{L}$ is the likelihood, $\boldsymbol{\theta} = \{d', c, \text{meta-}d', \text{meta-}c, \textbf{meta-c}_2^0, \textbf{meta-c}_2^1\}$, and $\boldsymbol{s}$, $\boldsymbol{r}$, and $\boldsymbol{c}$ vectors of observed stimuli, Type I responses, and Type II responses:

$$\mathcal{L}(\boldsymbol{\theta} \mid S = \boldsymbol{s}, R = \boldsymbol{r}, C = \boldsymbol{c}) = \prod_{n=1}^{N} P(R = r_n, C = c_n \mid S = s_n)$$

However, in this formulation the number of times each conditional probability is computed is proportional to the number of trials with the same stimulus, Type I response, and Type II response. As noted by Maniscalco and Lau (2014), it is often more efficient to use a multinomial likelihood over the aggregated data instead. Where $N_{s,r,c}$ is the number of trials with stimulus $S = s$, Type I response $R = r$, and Type II response $C = c$:

$$\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{N}) \propto \prod_{s,r,c} P(R = r, C = c \mid S = s)^{N_{s,r,c}}$$

For numerical stability, the `hmetad` package evaluates the model likelihood on the logarithmic scale. For a detailed derivation of the log likelihood used in the model fitting code, see Appendix.

Fixing the Type I criterion for Type II responses

In principle, one could treat meta-c as a free parameter reflecting the Type I criterion of the generative model for Type II responses. But to ease comparisons between meta-d' and d', in addition to comparisons across participants or experiments, it is instead customary to fix meta-c to be similar in some respect to the Type I criterion (c). Maniscalco and Lau (2014) advocate for two options in particular (Figure 3). In the ABSOLUTE model, meta-c is simply fixed to the Type I criterion (i.e., meta-c = c). This is the parameterization used in our earlier Hmeta-d toolbox and is the default in the `hmetad` package (Fleming, 2017). In the RELATIVE model, meta-c is fixed to be equal to the Type I criterion relative to the sensitivity (i.e., $\frac{\text{meta-}c}{\text{meta-}d'} = \frac{c}{d'}$), which can be achieved by setting meta-$c = \frac{\text{meta-}d'}{d'}c$. This is the parameterization used by Maniscalco and Lau (2014) and can be used in the `hmetad` package using the option `metac_absolute=FALSE`. Critically, Rausch et al. (2026) demonstrated that the choice between these two models is not arbitrary, as fitting data generated from the opposite generative model induces small but significant spurious dependencies of meta-d' on other model parameters.

Parameterization in the `hmetad` package

As described above, the likelihood of the meta-d' model is defined in terms of is parameters, $\boldsymbol{\theta} = \{d', c, \text{meta-}d', \text{meta-}c, \textbf{meta-c}_2^0, \textbf{meta-c}_2^1\}$. To enable efficient model fitting, the `hmetad` package uses an alternative parameterization that derives these parameters from transformed variables.

First, instead of fitting meta-d' directly, the `hmetad` package instead models the M-ratio, $M = \frac{\text{meta-}d'}{d'}$. To ensure that $0 \leq M \leq \infty$, the M-ratio is modeled with a logarithmic link function.[3] In this parameterization, one can compute meta-d' using the transformation meta-$d' = Md'$.

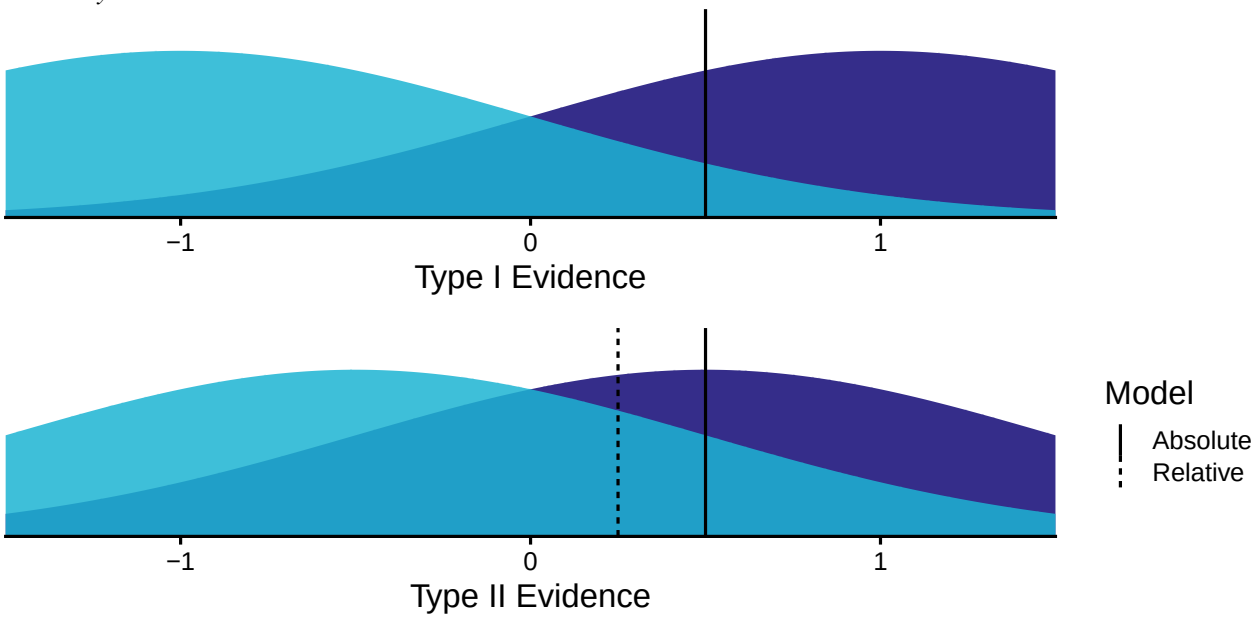
Second, in the original parameterization, the Type II criteria are ordinally constrained. Specifically, $\textbf{meta-c}_2^0$ is constrained to be decreasing and below meta-c, whereas $\textbf{meta-c}_2^1$ is constrained to be increasing and above meta-c:

$$\text{meta-}c_{2,K-1}^0 \leq \ldots \leq \text{meta-}c_{2,1}^0 \leq \text{meta-}c \leq \text{meta-}c_{2,1}^1 \leq \ldots \leq \text{meta-}c_{2,K-1}^1$$

[3] This constraint can be relaxed by setting the argument `allow_negative_values=TRUE`, which models M-ratio on its natural scale (i.e., using the identity link function).

Figure 3

Two options for fixing the Type I criterion for generating Type II responses. Under the ABSOLUTE model, the Type I criterion is shared between Type I and Type II observers. Under the RELATIVE model, the Type I criterion is scaled by the M-ratio so that the observers for the Type I and Type II decisions have equal response bias proportional to their sensitivity.



Previous implementations of the model either used constrained optimization (Maniscalco & Lau, 2014) or sorted the Type II criteria (Fleming, 2017) to meet this constraint. As a more flexible alternative, the `hmetad` package instead uses the positive ordered transform to model the *differences* between successive Type II criteria. Where K is the number of confidence levels, $K-1$ is the number of confidence criteria per response, we define $\delta\text{meta-}c^0_{2,k}$ and $\delta\text{meta-}c^1_{2,k}$ as the distances between Type II criteria for $R=0$ and $R=1$:

$$\delta\text{meta-}c^0_{2,k} = \begin{cases} \text{meta-}c - \text{meta-}c^0_{2,1} & \text{if } k = 1 \\ \text{meta-}c^0_{2,k-1} - \text{meta-}c^0_{2,k} & \text{if } 2 \le k \le K-1 \end{cases}$$

$$\delta\text{meta-}c^1_{2,k} = \begin{cases} \text{meta-}c^1_{2,1} - \text{meta-}c & \text{if } k = 1 \\ \text{meta-}c^1_{2,k} - \text{meta-}c^1_{2,k-1} & \text{if } 2 \le k \le K-1 \end{cases}$$

Under this parameterization, the actual Type II criteria are computed using the inverse transform:

$$\mathbf{meta\text{-}c^0_2} = \text{meta-}c - \text{cumulative_sum}\left(\delta\mathbf{meta\text{-}c^0_2}\right)$$

$$\mathbf{meta\text{-}c^1_2} = \text{meta-}c + \text{cumulative_sum}\left(\delta\mathbf{meta\text{-}c^1_2}\right)$$

As with the M-ratio, the distances $\delta\text{meta-}c^0_{2,k}$ and $\delta\text{meta-}c^1_{2,k}$ are modeled with a logarithmic link function to ensure that they are positive.

Including predictors

The primary benefit of the `hmetad` package is that it specifies the meta-d' model as a generalized linear model, allowing users to specify model parameters as a linear function of observed variables. For instance, where $\boldsymbol{X}_\varphi$ and β_φ

are the design matrix and coefficient vector for each parameter φ, the `hmetad` package uses the linear model:

$$\begin{aligned}
\log(M) &= \boldsymbol{X}_{M}\beta_{M} \\
d' &= \boldsymbol{X}_{\boldsymbol{d'}}\beta_{d'} \\
c &= \boldsymbol{X}_{\boldsymbol{c}}\beta_{c} \\
\log(\delta\text{meta-}c^0_{2,k}) &= \boldsymbol{X}_{\boldsymbol{\delta\text{meta-}c^0_{2,k}}}\beta_{\delta\text{meta-}c^0_{2,k}} \\
\log(\delta\text{meta-}c^1_{2,k}) &= \boldsymbol{X}_{\boldsymbol{\delta\text{meta-}c^1_{2,k}}}\beta_{\delta\text{meta-}c^1_{2,k}}
\end{aligned}$$

Importantly, the `hmetad` package allows for the design matrices to differ for each model parameter. As with any other `brms` family, it is also possible to include multilevel (or hierarchical) structure (Bürkner, 2018). Where $\boldsymbol{Z}_{\varphi}$ and γ_{φ} are the group-level design matrix and coefficient vector for each parameter φ:

$$\begin{aligned}
\log(M) &= \boldsymbol{X}_{M}\beta_{M} + \boldsymbol{Z}_{M}\gamma_{M} \\
d' &= \boldsymbol{X}_{\boldsymbol{d'}}\beta_{d'} + \boldsymbol{Z}_{\boldsymbol{d'}}\gamma_{d'} \\
c &= \boldsymbol{X}_{\boldsymbol{c}}\beta_{c} + \boldsymbol{Z}_{\boldsymbol{c}}\gamma_{c} \\
\log(\delta\text{meta-}c^0_{2,k}) &= \boldsymbol{X}_{\boldsymbol{\delta\text{meta-}c^0_{2,k}}}\beta_{\delta\text{meta-}c^0_{2,k}} + \boldsymbol{Z}_{\boldsymbol{\delta\text{meta-}c^0_{2,k}}}\gamma_{\delta\text{meta-}c^0_{2,k}} \\
\log(\delta\text{meta-}c^1_{2,k}) &= \boldsymbol{X}_{\boldsymbol{\delta\text{meta-}c^1_{2,k}}}\beta_{\delta\text{meta-}c^1_{2,k}} + \boldsymbol{Z}_{\boldsymbol{\delta\text{meta-}c^1_{2,k}}}\gamma_{\delta\text{meta-}c^1_{2,k}}
\end{aligned}$$

Although a full description of all of the different kinds of predictors allowed in `brms` is outside the scope of this article, `brms` also allows for non-linear terms, monotonic predictors, autocorrelation terms, and predictors with measurement error (Bürkner, 2017, 2018; Bürkner & Charpentier, 2020).

Fitting the meta-d' model in R

As a practical demonstration of how to use the `hmetad` package to fit the meta-d' model in R, here we reanalyze data from Rouault et al. (2018) Experiment 2. In this experiment, 497 participants (after exclusion) completed a two-alternative forced-choice perceptual decision-making task in which they were asked to decide which of two boxes contained more dots. Participants rated their confidence in their perceptual decisions on a six-point scale from "guessing" to "certainly correct". Participants' performance on the perceptual task was controlled using a two-down one-up staircase on the logarithm of the difference in the number of dots between the two boxes. Finally, participants completed a number of self-report psychiatric questionnaires, which were aggregated into three factors (i.e., anxious-depression, compulsive behavior and intrusive thought, and social withdrawal) using factor analysis (for more information, see Rouault et al., 2018). For simplicity, we will focus on the effects of the anxious-depression factor on metacognition, returning to the effects of the other two factors afterwards.

In their analysis, Rouault et al. (2018) correlated these three factors with independent single-participant estimates of M-ratio and mean confidence. This approach has two main drawbacks: (a) participant-level estimates of metacognitive factors are noisy and thus benefit strongly from the partial pooling present in multilevel models (Fleming, 2017; Rahnev, 2025), and (b) correlating participant-level estimates with mental health factors ignores uncertainty in participant-level estimates, as well as potential correlations in uncertainty, leading to miscalibrated statistical inferences (Moutoussis et al., 2018; Singmann & Kellen, 2019). Here, we overcome these limitations by directly modeling the association of anxious-depression with the parameters of the meta-d' model, where participant-level parameters are estimated hierarchically.

Setup

Installation

To install the latest release of the `hmetad` package from CRAN, one can use the command `install.packages("hmetad")`. Alternatively, the development version of `hmetad` can be installed using `pak::pak("metacoglab/hmetad")`.

The `hmetad` package depends on the `brms` package, which in turn uses either the `Rstan` or `cmdstanr` interface to the probabilistic programming language Stan. Installing `hmetad` will install `Rstan` automatically, however it is also recommended to install the `cmdstanr` package, since it produces more optimized code for most applications. In

either case, both interfaces require a working C++ compiler specific to your computer's operating system. In particular, Windows users must install RTools (https://cran.r-project.org/bin/windows/Rtools/). For detailed information on installation, please see https://mc-stan.org/install/.

Once installed, the `hmetad` package can be loaded using the command `library(hmetad)`.

Data preparation

Most experimental software is designed to save trial-level tabular data, with one row for each individual trial and columns for each independent or dependent variable. The raw data provided by Rouault et al. (2018) also takes this form:

```
d <- read_csv("rouault_2018_exp2.csv") |>
  mutate(
    across(stimulus_left:confidence, as.integer),
    across(c(participant, gender_female), factor)
  )
```

```
# A tibble: 103,322 x 9
   participant stimulus_left response_left confidence   age gender_female
   <fct>               <int>         <int>      <int> <dbl> <fct>
 1 1015841                 1             1          4    32 0
 2 1015841                 0             0          4    32 0
 3 1015841                 0             0          5    32 0
 4 1015841                 1             1          4    32 0
 5 1015841                 0             1          2    32 0
 6 1015841                 0             0          3    32 0
 7 1015841                 1             1          5    32 0
 8 1015841                 1             1          4    32 0
 9 1015841                 0             0          3    32 0
10 1015841                 0             0          4    32 0
# i 103,312 more rows
# i 3 more variables: anxious_depression <dbl>, compulsivity <dbl>,
#   social_withdrawal <dbl>
```

As we can see, this is quite a large data set with 103322 observations, where `participant` is the participant identifier, `stimulus_left` indicates whether the box with more dots was on the left side, `response_left` indicates whether the participant reported that the left box had more dots, `confidence` indicates their confidence rating, `age` and `gender_female` are demographic variables, and `anxious_depression`, `compulsivity`, and `social_withdrawal` are the psychiatric indices generated from factor analysis (see Rouault et al., 2018 for more information).

To increase the efficiency of model fitting, `hmetad` fits the meta-d' model on aggregated data rather than this trial-level data. In practice, `hmetad` automatically generates this aggregated data set internally based on the provided model formula, so users are not required to aggregate their own data. Nevertheless, if we would like, we can manually aggregate the data using the function `aggregate_metad(d, ...)`, where `...` are grouping columns by which to aggregate. By default, this function assumes that `d` has a column named `stimulus` and either separate columns named `response` and `confidence` or a single `joint_response` column. Since our data has the more informative column names `stimulus_left` and `response_left`, we can indicate this using the optional arguments `.stimulus` and `.response`. For instance, we can aggregate the data separately for each participant like so:

```
aggregate_metad(
  d, participant,
  .stimulus = "stimulus_left", .response = "response_left"
)
```

```
# A tibble: 497 x 4
   participant   N_0   N_1 N[,"N_0_1"] [,"N_0_2"] [,"N_0_3"] [,"N_0_4"]
```

```
   <fct>       <int> <int>       <int>      <int>      <int>      <int>
 1 1015841        75   134           0         17         30          5
 2 1030622       103   104           0          0          1          1
 3 1103437       108   100           8         12         14         13
 4 1144714       118    88           0         20         45         13
 5 1149809        90   117          18         24         24         13
 6 1171324       129    80           7         26         27         18
 7 1202074       104   103           0          1         12         30
 8 1211330       111    78          20         11          7         20
 9 1218685        90   119           0          0          3         48
10 1234928        90   118           1         15         24          6
# i 487 more rows
# i 1 more variable: N[5:24] <int>
```

Instead of one row per trial, this aggregated data set has only 497 rows (one row per participant). Here, the `N_0` and `N_1` columns represent the number of trials where `stimulus_left==0` and `stimulus_left==1`, respectively, and the `N` column is a matrix containing the joint distribution of stimuli, Type I responses, and Type II responses. Its sub-columns are named with the pattern `N_<stimulus>_<joint_response>`, such that `N_0_1` contains the number of trials with `stimulus==0` and `joint_response==1` (i.e., `response_left==0` and `confidence==6`) and `N_1_7` contains the number of trials with `stimulus==1` and `joint_response==7` (i.e., `response_left==1` and `confidence==1`).

We are free to add any other grouping columns for data aggregation. Since we will be using anxious-depression as a predictor, for example, we can include it like so:

```
aggregate_metad(
  d, participant, anxious_depression,
  .stimulus = "stimulus_left", .response = "response_left"
)
```

```
# A tibble: 497 x 5
   participant anxious_depression   N_0   N_1 N[,"N_0_1"] [,"N_0_2"] [,"N_0_3"]
   <fct>                    <dbl> <int> <int>       <int>      <int>      <int>
 1 1015841                 -0.126    75   134           0         17         30
 2 1030622                 -1.31    103   104           0          0          1
 3 1103437                 -1.01    108   100           8         12         14
 4 1144714                  0.631   118    88           0         20         45
 5 1149809                 -0.759    90   117          18         24         24
 6 1171324                  0.164   129    80           7         26         27
 7 1202074                  1.49    104   103           0          1         12
 8 1211330                 -1.27    111    78          20         11          7
 9 1218685                 -0.308    90   119           0          0          3
10 1234928                  0.802    90   118           1         15         24
# i 487 more rows
# i 1 more variable: N[4:24] <int>
```

Empirical quantities. The `aggregate_metad` function converts the data into a format that is useful for model fitting, but this format is rather inconvenient for other purposes like plotting and computation of response probabilities. So, the `hmetad` package provides several functions to compute data-driven measures relevant to metacognition researchers, all of which share the same basic interface as `aggregate_metad`. To compute the Type II response probabilities per participant, for instance, we can use the function `type2_probabilities`:

```
type2_probabilities(
  d, participant, anxious_depression,
  .stimulus = "stimulus_left", .response = "response_left"
)
```

```
# A tibble: 11,928 x 8
# Groups:   participant, anxious_depression, stimulus_left, response_left
#   [1,988]
   participant anxious_depression stimulus_left response_left confidence
   <fct>                    <dbl>         <int>         <int>      <int>
 1 1015841                -0.126             0             0          1
 2 1015841                -0.126             0             0          2
 3 1015841                -0.126             0             0          3
 4 1015841                -0.126             0             0          4
 5 1015841                -0.126             0             0          5
 6 1015841                -0.126             0             0          6
 7 1015841                -0.126             0             1          1
 8 1015841                -0.126             0             1          2
 9 1015841                -0.126             0             1          3
10 1015841                -0.126             0             1          4
# i 11,918 more rows
# i 3 more variables: joint_response <int>, n <int>, p <dbl>
```

We can compute the area under the Type II ROC (AUROC2) similarly:

```
auroc2(
  d, participant, anxious_depression,
  .stimulus = "stimulus_left", .response = "response_left"
)
```

```
# A tibble: 994 x 4
# Groups:   participant, anxious_depression [497]
   participant anxious_depression response_left auroc2
   <fct>                    <dbl>         <int>  <dbl>
 1 1015841                -0.126             0  0.630
 2 1015841                -0.126             1  0.712
 3 1030622                -1.31              0  0.568
 4 1030622                -1.31              1  0.610
 5 1103437                -1.01              0  0.671
 6 1103437                -1.01              1  0.861
 7 1144714                 0.631             0  0.629
 8 1144714                 0.631             1  0.609
 9 1149809                -0.759             0  0.762
10 1149809                -0.759             1  0.593
# i 984 more rows
```

As we demonstrate below, each of these functions has another corresponding function for extracting model estimates of the same quantities to facilitate comparison between the data and model predictions.

Model specification

Given a data set, the next step is to specify a model that properly accounts for the nested structure of the data. Here we focus on specifying a model for the data collected by Rouault et al. (2018) as an example. A full characterization of available model designs is outside the scope of this article; we refer readers to (Bürkner, 2018; Singmann & Kellen, 2019) for more information.

As can be seen from the aggregated data, Rouault et al. (2018) used a mixed design with repeated trials per participant, but with anxious-depression as a between-participant predictor. To account for this structure, we want a model that hierarchically estimates separate parameters for each participant with population-level effects of anxious-depression (Barr, 2013). In the `hmetad` package, we can specify this structure as an R formula as used in the `lme4` and `brms` packages.

First, to verify which parameters our model contains, it is helpful to view the distribution family for our data:

```
metad(n_distinct(d$confidence))
```

```
Custom family: metad__6__normal__absolute__multinomial
Link function: log
Parameters: mu, dprime, c, metac2zero1diff, metac2zero2diff,
  metac2zero3diff, metac2zero4diff, metac2zero5diff, metac2one1diff,
  metac2one2diff, metac2one3diff, metac2one4diff, metac2one5diff
```

This output tells us a few key pieces of information. Looking at the name of the model family, we see that our model has 6 confidence levels and uses the normal distribution for the signal detection model, the ABSOLUTE variant of the meta-d' model, and the multinomial likelihood (over aggregated data). This output also lists the model parameters over which we can specify `brms` formulas. The primary model parameter is the M-ratio, which for technical reasons, `brms` requires to be called `mu`. As the model uses the `log` link function, M-ratio is estimated on the logarithmic scale to prevent negative estimates of M-ratio[4]. `dprime` and `c` are the Type I sensitivity and response criterion estimated on the identity scale. Finally, we have a long list of parameters which define the Type II criteria. These parameters are named using the syntax `metac2<response><confidence>diff`, where `<response>` is the Type I response (either `zero` or `one`) and `<confidence>` is any integer from `1` to `5` (one less than the total number of confidence levels).

Now that we know the names of all of the parameters for our model, we can specify a model formula for each parameter using the function `bf` from the `brms` package:

```
formula <- bf(
  N ~ anxious_depression + (1 | participant),
  dprime ~ anxious_depression + (1 | participant),
  c ~ anxious_depression + (1 | participant),
  metac2zero1diff ~ anxious_depression + (1 | p | participant),
  metac2zero2diff ~ anxious_depression + (1 | p | participant),
  metac2zero3diff ~ anxious_depression + (1 | p | participant),
  metac2zero4diff ~ anxious_depression + (1 | p | participant),
  metac2zero5diff ~ anxious_depression + (1 | p | participant),
  metac2one1diff ~ anxious_depression + (1 | p | participant),
  metac2one2diff ~ anxious_depression + (1 | p | participant),
  metac2one3diff ~ anxious_depression + (1 | p | participant),
  metac2one4diff ~ anxious_depression + (1 | p | participant),
  metac2one5diff ~ anxious_depression + (1 | p | participant)
)
```

This formula has a few notable features. First, to specify the formula for M-ratio, we use the response variable `N` (which is the name of the column of aggregated responses from `aggregate_metad`). Second, for each parameter we include `anxious_depression` as a between-participant predictor in the model. Third, we include hierarchical estimates of all coefficients for each participant with the syntax `(1 | participant)` and `(1 |p| participant)`. This syntax results in random intercepts for each participant, meaning that this model will capture participant-level deviations from model estimates.[5] Here, we use `(1 |p| participant)` for the confidence criteria to additionally model participant-level correlations (since it is expected that participants with one confidence criterion set high will likely have others set high as well). As we are using the same design for all model parameters, the formula above has a lot of repeated code. To make the code more readable, we can specify formulas for multiple model parameters at the same time using the `+` operator (except for M-ratio, which requires its own formula):

```
formula <- bf(
  N ~ anxious_depression + (1 | participant),
  dprime + c ~ anxious_depression + (1 | participant),
  metac2zero1diff + metac2zero2diff + metac2zero3diff +
```

[4]This constraint can be relaxed by setting the argument `allow_negative_values = TRUE`.

[5]Note that if we had within-participant predictors, it would be appropriate to include those in this formula.

```
    metac2zero4diff + metac2zero5diff +
    metac2one1diff + metac2one2diff + metac2one3diff +
    metac2one4diff + metac2one5diff ~
    anxious_depression + (1 | p | participant)
)
```

Prior elicitation

Before fitting the meta-d' model, it is necessary to specify priors over model parameters using multiple calls to the `set_prior` function (or one of its variants) from the `brms` package. Generally, prior specification will depend on a mixture of domain expertise about the model, the data being modeled, and the population being studied. The general goal is to choose priors that match our broad assumptions about the patterns expected in the data. Critically, these expectations are typically independent of our hypothesis test(s) of interest—in this case, the link between anxious-depression level and metacognition. For instance, in the case of the meta-d' model, we should select priors on confidence criteria that approximate a reasonable distribution of confidence ratings when averaged across conditions of an experiment, or across individuals in a population. Appropriate prior specification is important to enable efficient and accurate model fits, just as selecting appropriate initializations and bounds on parameters is important for any constrained optimization problem. Because prior specification is a somewhat subjective process, we recommend the use of prior predictive checks—the simulation of model parameters and predictions from the prior distribution—as a standard part of the Bayesian workflow (Gelman et al., 2026; Schad et al., 2021).

Here we chose to use normal priors over the model coefficients, half-normal priors for the between-participant standard deviations, and LKJ priors for between-participant correlations:

```
prior <- set_prior("normal(0, .33)", class = "Intercept") +
  set_prior("normal(1.11, .1)", class = "Intercept", dpar = "dprime") +
  set_prior("normal(0, .1)", class = "Intercept", dpar = "c") +
  set_prior("normal(0, 1)", class = "Intercept", dpar = metac2_parameters(K = 6)) +
  set_prior("normal(0, 0.25)", class = "b") +
  set_prior("normal(0, 0.25)", class = "b", dpar = c("dprime", "c")) +
  set_prior("normal(0, 0.25)", class = "b", dpar = metac2_parameters(K = 6)) +
  set_prior("normal(0, 0.5)", class = "sd") +
  set_prior("normal(0, 0.33)", class = "sd", dpar = c("dprime", "c")) +
  set_prior("normal(0, 1)", class = "sd", dpar = metac2_parameters(K = 6)) +
  set_prior("lkj(2)", class = "cor")
```

Because we are modeling each parameter hierarchically, each parameter has a separate prior for its population-level intercepts (with the argument `class = "Intercept"`), population-level coefficients (with the argument `class = "b"`) and for the standard deviation of between-participant differences (with the argument `class = "sd"`). These priors have been selected to match our expectations for the particular design used by Rouault et al. (2018). In particular, for perceptual tasks we generally expect mean M-ratio to lie somewhere between `0.5` and `2.0`, with participant-level M-ratios having slightly more variance. Next, because this experiment (a) used a two-alternative forced-choice design to reduce response biases and (b) included a two-down one-up staircase to fix task performance to $\sim 71\%$ for all participants, we have a strong expectation that $c \approx 0$ and $d' \approx 2\Phi^{-1}(.71) \approx 1.11$. To account for this, we set reasonably informative priors over both parameters, as well as priors over the between-participant standard deviations that encourage small differences in first-order performance between participants.

The priors for the confidence criteria are the most difficult to set, because they are on an unintuitive scale (i.e., log distances between successive criteria) and because the optimal criterion setting will depend on the number of confidence levels available to the participant (among other factors, see Maniscalco et al., 2025). As a starting point, here was have arbitrarily chosen standard normal priors for these parameters. Rather than relying on such choices in our analysis, however, here we will iteratively refine our prior distribution through the use of prior predictive checks. As described above, these prior predictive checks are carried out before evaluating our hypothesis test about the relationship between anxious-depression level, to avoid biasing our choice of priors. So, to check the implications of this prior distribution, we can fit a prior-only model by calling the function `fit_metad` with the argument `sample_prior = "only"`:

```
m.prior <- fit_metad(
  formula, d,
  .stimulus = "stimulus_left", .response = "response_left",
  file = "_models/prior_stdnormal.rds", prior = prior, sample_prior = "only",
  cores = 4, init = 0
)
```

Here we have chosen to save our prior model into a file called `prior_stdnormal.rds`, to run Stan in parallel across four cores, and to initialize all parameters to zero. Like any other `brms` object, we can view helpful summaries of our model using the `summary` and `plot` functions. Since this model has a large number of parameters, however, it is easier to check the implications of this prior distribution by plotting out distributions of model parameters and other implied quantities.

The `hmetad` package has a range of functions for extracting such model quantities, all of which contain either `_draws` or `_rvars` following conventions from the `tidybayes` package (Kay, 2024). Each of these functions takes as input a `brms` model and a new dataset from which to compute predictions. To check our priors for the group-level parameters, we can first display the prior distributions of model parameters using the function `linpred_rvars_metad` over a sample dataset where anxious-depression is set to zero (i.e., its mean value) and `tidybayes::median_qi` to display prior medians and 95% credible intervals:

```
newdata <- tibble(anxious_depression = 0)

linpred_rvars_metad(m.prior, newdata, re_formula = NA, pivot_longer = TRUE) |>
  median_qi(.value)
```

```
# A tibble: 15 x 9
   anxious_depression  .row .variable      .value   .lower .upper .width .point
                <dbl> <int> <chr>           <dbl>    <dbl>  <dbl>  <dbl> <chr>
 1                  0     1 M            1.00       0.513   1.94    0.95 median
 2                  0     1 dprime       1.11       0.921   1.31    0.95 median
 3                  0     1 c           -0.000103  -0.205   0.201   0.95 median
 4                  0     1 meta_dprime  1.11       0.554   2.19    0.95 median
 5                  0     1 meta_c      -0.000103  -0.205   0.201   0.95 median
 6                  0     1 meta_c2_0_1 -0.989     -7.06   -0.111   0.95 median
 7                  0     1 meta_c2_0_2 -2.44     -11.1    -0.577   0.95 median
 8                  0     1 meta_c2_0_3 -3.98     -14.6    -1.18    0.95 median
 9                  0     1 meta_c2_0_4 -5.47     -17.4    -1.88    0.95 median
10                  0     1 meta_c2_0_5 -7.02     -20.4    -2.69    0.95 median
11                  0     1 meta_c2_1_1  0.998      0.1000  6.92    0.95 median
12                  0     1 meta_c2_1_2  2.44       0.545  11.8     0.95 median
13                  0     1 meta_c2_1_3  3.95       1.11   14.8     0.95 median
14                  0     1 meta_c2_1_4  5.48       1.81   18.4     0.95 median
15                  0     1 meta_c2_1_5  7.07       2.74   20.6     0.95 median
# i 1 more variable: .interval <chr>
```

Just by looking at the prior summary, we can see that this prior places reasonable bounds for mean M-ratio, d', and c. We can also see that our priors for the mean Type II criteria are ordered appropriately with a substantial degree of uncertainty, though it is still not immediately clear whether these values are plausible. Note that, in this code, participant-level differences are ignored by setting `re_formula = NA` (otherwise, `newdata` must contain a column called `participant` containing the ID of the participant being simulated).

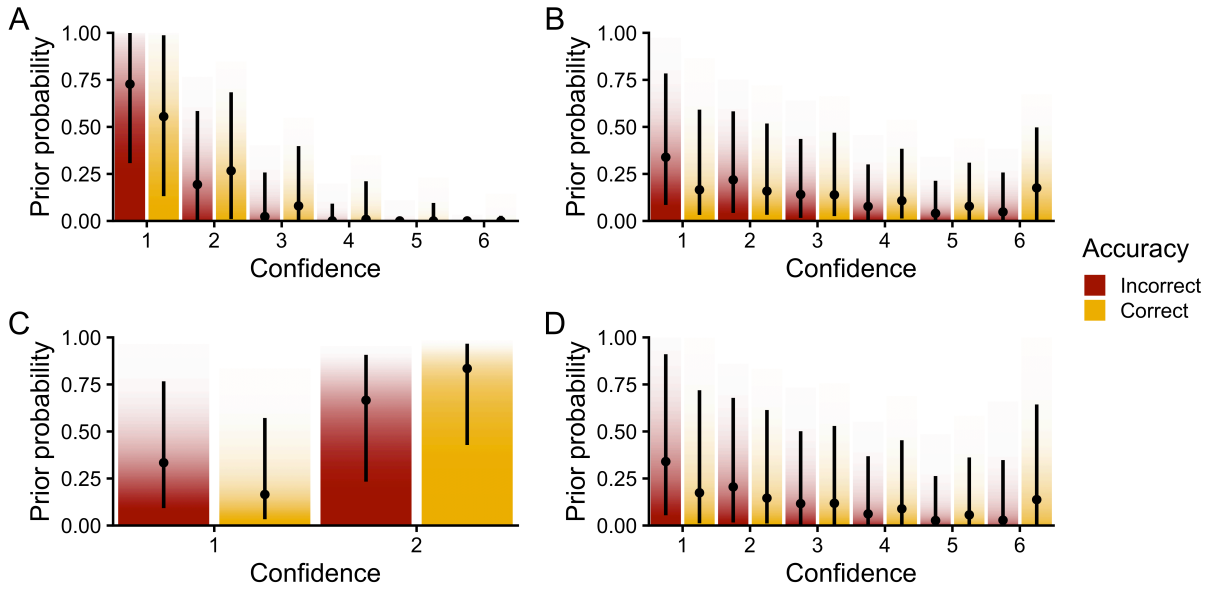
As a further check, we can compute the implied Type II response probabilities using the analogous function `type2_rvars(m.prior, newdata, re_formula = NA, by_correct = TRUE)` (see Figure 4A). From this figure, it is clear that the standard normal priors for the confidence criteria place far too much probability on low-confidence responses, such that the maximum confidence level has almost no prior probability. To fix this issue, we iteratively adjusted the prior mean until the prior predictions appear suitable. This process revealed that a prior mean

of -1.33 generated prior predictions that put roughly equal probability on each confidence level for correct responses with a slight preference for low confidence for incorrect responses, which is a reasonable starting point Figure 4B.

Critically, it is necessary to tune the prior for the Type II criteria through iterative refinement and visualization, because models with different number of confidence criteria require different parameter values to produce roughly uniform confidence responding. As an example, if we use the same prior for a hypothetical dataset with only two confidence levels, the resulting prior predictions strongly favor high-confidence responding (Figure 4C). In this scenario, it would be sensible to increase the prior mean of the confidence criteria until the prior predictions more closely resembled our expectations.

Figure 4

Prior predictions of Type II response probabilities. (A) *Group-level prior predictions with anxious-depression at the mean, using a standard normal prior for the distances between confidence criteria on the logarithmic scale. These predictions are highly skewed to favor low-confidence responses, suggesting that these confidence criteria are too far apart.* (B) *Group-level prior predictions using a revised prior with the mean distance between criteria reduced to -1.33 on the logarithmic scale. Prior predictions are now mostly uniform, with a slight preference for low confidence following incorrect responses.* (C) *Group-level prior predictions using the same prior as in* (B)*, only for a dataset with two confidence levels. Unlike in* (B)*, this prior favors high confidence ratings and should likely be adjusted before use.* (D) *Participant-level prior predictions using the same prior as in* (B) *for an example participant.*



Note. Points and intervals indicate prior medians and 95% credible intervals. Smoothed bars depict complementary cumulative distribution function of prior distributions.

Having settled on a prior for the group-level confidence criteria, we next visualize priors for participant-level standard deviations by simulating just a single participant (since all participants are equivalent *a priori*):

```
newdata |>
  mutate(participant = first(d$participant)) |>
  add_linpred_rvars_metad(m.prior, pivot_longer = TRUE) |>
  median_qi(.value)
```

```
# A tibble: 15 x 10
   anxious_depression participant  .row .variable     .value    .lower   .upper
                <dbl> <fct>       <int> <chr>          <dbl>     <dbl>    <dbl>
 1                  0 1015841         1 M            0.995     0.309    3.29
 2                  0 1015841         1 dprime       1.11      0.419    1.84
```

```
 3                  0 1015841         1 c           -0.00194  -0.674    0.744
 4                  0 1015841         1 meta_dprime  1.10      0.200    3.93
 5                  0 1015841         1 meta_c      -0.00194  -0.674    0.744
 6                  0 1015841         1 meta_c2_0_1 -0.326    -4.35     0.497
 7                  0 1015841         1 meta_c2_0_2 -0.755    -8.43     0.206
 8                  0 1015841         1 meta_c2_0_3 -1.25    -11.8     -0.00590
 9                  0 1015841         1 meta_c2_0_4 -1.79    -16.7     -0.243
10                  0 1015841         1 meta_c2_0_5 -2.34    -20.1     -0.467
11                  0 1015841         1 meta_c2_1_1  0.333    -0.452    4.05
12                  0 1015841         1 meta_c2_1_2  0.768    -0.225    7.37
13                  0 1015841         1 meta_c2_1_3  1.27      0.00714 11.0
14                  0 1015841         1 meta_c2_1_4  1.80      0.250   16.6
15                  0 1015841         1 meta_c2_1_5  2.35      0.460   19.0
# i 3 more variables: .width <dbl>, .point <chr>, .interval <chr>
```

The resulting probabilities are plotted in Figure 4D. Note that participant-level priors will always be centered around the population means, only with greater uncertainty. Just as with the population-level parameters, we can use visualizations like this to iteratively elicit prior distributions that reflect our beliefs about reasonable differences between participants. For instance, we can increase the participant-level standard deviations to allow a wider range of behaviors or alternatively decrease the participant-level standard deviations to encourage stronger similarity between participants. This strategy is also useful for setting priors for model coefficients, for example to visualize differences in prior distributions at different levels of anxious-depression.

We end this section by highlighting two key points. First, there can be reasonable disagreement about the "best" prior to use in any given situation; as a result, one's choice of prior need only be scientifically justifiable in the context of one's particular study (McElreath et al., 2016). Moreover, so long as the prior is not exceedingly informative, its influence on the resulting posterior should be small (Kruschke & Liddell, 2018). Second, **prior revision should always occur before looking at the data**, since refining a prior after observing the data is a form of double-dipping tantamount to Bayesian *p*-hacking. Incorporating thorough prior predictive checks into a standard workflow can help prevent such practices by ensuring that one's model is validated before fitting it to data (Gelman et al., 2026).

Model fitting

Now that we have verified that our prior model is sufficient, we can fit the model to our data by calling the `update` function on our prior model with the argument `sample_prior = "no"` (alternatively, we could call `fit_metad` function again, except this time removing the argument `sample_prior = "only"`). Note that the time it takes to fit a model is mostly dependent on the number of rows in the aggregated dataset (i.e., participants and within-participant conditions), not the number of trials. Most simple models should fit within a few minutes; however models with many predictors or models fit to large datasets such as this one may take over an hour. In such cases, we recommend using a smaller subset of the data during model development.

```
m <- update(m.prior, file = "_models/fit.rds", sample_prior = "no", cores = 4)
```

Like any `brms` model, we can obtain numerical and graphical summaries of the posterior distribution using `summary(m)` and `plot(m)`. Before inspecting the fitted model, however, it is necessary to ensure that the model has converged properly.

Model convergence

There several possible warning messages that one might encounter during model fitting. Some warnings, such as warnings about effective sample size, do not indicate serious problems with the model itself and can usually be resolved with minor adjustments (e.g., running the model for a larger number of iterations). But most of the diagnostics checked by Stan *do* indicate problems with the model that must be resolved before interpreting the resulting posterior. Here we will focus on one common issue resulting from hierarchical models: divergent transitions. For a detailed guide on resolving other model fitting issues, we recommend the Stan guide on convergence warnings (https://mc-stan.org/learn-stan/diagnostics-warnings.html).

In this case, R outputs the following warnings:

```
Parts of the model have not converged (some Rhats are > 1.05). Be careful when
analysing the results! We recommend running more iterations and/or setting
stronger priors.
There were 4000 divergent transitions after warmup.
Increasing adapt_delta above 0.8 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

When `brms` outputs warnings that Stan has encountered divergent transitions, this is an indicator that the model's posterior distribution has not been fully explored, meaning that the posterior samples under-represent the true posterior (Betancourt, 2017).[6] If there are only a small number of divergent transitions, it may be sufficient to adjust Stan's settings to enable more fine-grained sampling using the argument `control = list(adapt_delta = <adapt_delta>)` (indeed, this is the default recommendation accompanied by warnings about divergent transitions). However, this approach will significantly slow down model fitting and is not guaranteed to resolve the issue.

The first potential solution to this problem is to set more informative priors to help bound difficult posterior geometries (Betancourt, 2017). By default, `brms` uses Student-t priors over intercepts and participant-level standard deviations and uninformative (i.e., flat) priors over all other parameters. Such uninformative priors not only are unlikely to represent one's beliefs about the model parameters, but they also put substantial probability mass over implausible values (e.g., a flat prior over M-ratio makes the values $M = 1$ and $M = 100000$ equally likely). To verify that priors have been set properly, it is helpful to check the output of `prior_summary(m)`. As described above, here we have chosen moderately informative priors based on knowledge about the experimental design. These constraints already help jointly identify the model parameters, so in this case constraining the prior further is unlikely to resolve convergence issues.

Another potential solution is to investigate which parameters exhibit problematic posterior geometry and eliminate them from the model (Gabry et al., 2019). In the context of the meta-d' model, a common source of such problems is that parameters are poorly identified when participants exhibit aberrant behavior. For instance, if a participant only responds "present" (or, "right"), it is impossible to estimate confidence criteria for the opposite response. As another example, it is difficult to estimate M-ratio when a participant has chance-level task performance (i.e., $d' = 0$), because dividing by zero creates infinite values of M-ratio. In our case, however, the divergent transitions are likely due to extreme confidence responding: there are many participants in our data who only use one or two (of six) confidence levels, leaving many cells with zero counts in the aggregated data. Such issues are somewhat mitigated in a multilevel model such as the one we present here, since participant-level estimates are informed by the group-level means. However, if there are enough empty cells, or if the group-level estimates themselves are under-powered, participant-level estimates of the confidence thresholds may still be poorly identified or exhibit degenerate posterior geometries.

In such cases, it may be necessary to either reduce the number of confidence ratings through binning or to exclude participants with highly skewed response patterns, so long as exclusion does not introduce post-treatment bias or other kinds of selection effects (Rohrer, 2024). Another alternative is to perform analyses collapsing across participants, with the caveat that collapsed estimates are less likely to generalize across the population (Yarkoni, 2020). Many researchers have also used data padding (or edge correction)—imputing artificial trials into the dataset to avoid zero counts—to mitigate similar identifiability issues. However, padding is generally unadvisable as it introduces biases into the estimates of Type I sensitivity, which in turn biases estimates of M-ratio (Fleming, 2017; Hautus, 1995). In any case, it is usually necessary to perform robustness checks to ensure the key pattern of results is not sensitive to the data cleaning process.

Such issues can ideally be mitigated prior to data collection using careful experimental designs which encourage participants to (a) perform above chance, (b) use both Type I responses equally, and (c) make full use of the confidence scale. In the absence of these strategies, here we resolve our model convergence issues by binning the confidence data from six to three levels using the same cut-points for all participants[7] and by excluding 44 (~9%) participants that used less than 2/3 of possible joint response options.

[6] Divergent transitions are a diagnostic specific to Hamiltonian Monte Carlo, the model-fitting algorithm used by Stan. Models previously fit using JAGS within the Hmeta-d toolbox may exhibit such warnings when fit with the `hmetad` package. Because JAGS does not provide such diagnostics, and because a lack of diagnostic warnings does not guarantee that a model has converged, this means that the posterior approximations obtained by JAGS may also be jeopardized.

[7] Alternatively, if metacognitive bias is not of scientific interest, one can bin confidence ratings separately by participant by grouping the data prior to binning.

```r
## bin data and exclude participants with >1/3 empty cells
d <- d |>
  mutate(confidence = cut(confidence, breaks = c(0.5, 3.5, 4.5, 6.5), labels = FALSE))
exclude_ids <- joint_probabilities(
  d, participant,
  .stimulus = "stimulus_left", .response = "response_left"
) |>
  group_by(participant) |>
  summarize(n_zeros = sum(n == 0)) |>
  filter(n_zeros >= 4) |>
  pull(participant)
d <- d |>
  filter_out(participant %in% exclude_ids) |>
  mutate(participant = factor(participant))


## re-fit model on cleaned data
formula <- bf(
  N ~ anxious_depression + (1 | participant),
  dprime + c ~ anxious_depression + (1 | participant),
  metac2zero1diff + metac2zero2diff +
    metac2one1diff + metac2one2diff ~
    anxious_depression + (1 | p | participant)
)

prior <- set_prior("normal(0, .33)", class = "Intercept") +
  set_prior("normal(1.11, .1)", class = "Intercept", dpar = "dprime") +
  set_prior("normal(0, .1)", class = "Intercept", dpar = "c") +
  set_prior("normal(-1, 1)", class = "Intercept", dpar = metac2_parameters(K = 3)) +
  set_prior("normal(0, 0.25)", class = "b") +
  set_prior("normal(0, 0.25)", class = "b", dpar = c("dprime", "c")) +
  set_prior("normal(0, 0.25)", class = "b", dpar = metac2_parameters(K = 3)) +
  set_prior("normal(0, 0.5)", class = "sd") +
  set_prior("normal(0, 0.33)", class = "sd", dpar = c("dprime", "c")) +
  set_prior("normal(0, 1)", class = "sd", dpar = metac2_parameters(K = 3))

m.prior <- fit_metad(
  formula, d,
  .stimulus = "stimulus_left", .response = "response_left",
  file = "_models/prior_binned.rds",
  prior = prior, sample_prior = "only",
  iter = 6000, warmup = 1000,
  chains = 4, cores = 4, init = 0
)

m <- update(m.prior,
  file = "_models/fit_binned.rds", cores = 4,
  sample_prior = "no", control = list(adapt_delta = 0.95)
)
```

Model validation and interpretation

In this section, we demonstrate how to assess model fit and interpret model parameters using posterior retrodictive checks with the `hmetad` package. A posterior retrodictive check compares visualizations of our empirical data

with the same visualizations computed from data simulated from the fitted model parameters. A close match indicates that the model is adequately capturing the qualitative features of the data. That is, we would like to know whether the model makes predictions of Type I and Type II responses that resemble the responses in the data. Having already completed a thorough prior predictive check, the posterior retrodictive check is straightforward: simply generate the same estimates and visualizations using the posterior distribution, only this time overlaying the data for comparison.

As a preliminary check, we can obtain the joint Type I/Type II response probabilities averaged across all participants. Here we focus on joint probabilities (rather than simply Type II probabilities) because they reflect differences in both Type I and Type II parameters.

```
draws.epred <- d |>
  distinct(participant, anxious_depression) |>
  add_epred_rvars_metad(m, .stimulus = "stimulus_left", .response = "response_left") |>
  group_by(stimulus_left, joint_response) |>
  summarize(.epred = rvar_mean(.epred))
draws.epred
```

```
# A tibble: 12 x 3
# Groups:   stimulus_left [2]
   stimulus_left joint_response           .epred
           <int>          <int>       <rvar[1d]>
 1             0              1  0.281 ± 0.00176
 2             0              2  0.205 ± 0.00158
 3             0              3  0.217 ± 0.00160
 4             0              4  0.150 ± 0.00144
 5             0              5  0.079 ± 0.00087
 6             0              6  0.068 ± 0.00090
 7             1              1  0.052 ± 0.00075
 8             1              2  0.068 ± 0.00081
 9             1              3  0.120 ± 0.00129
10             1              4  0.229 ± 0.00162
11             1              5  0.207 ± 0.00156
12             1              6  0.324 ± 0.00181
```

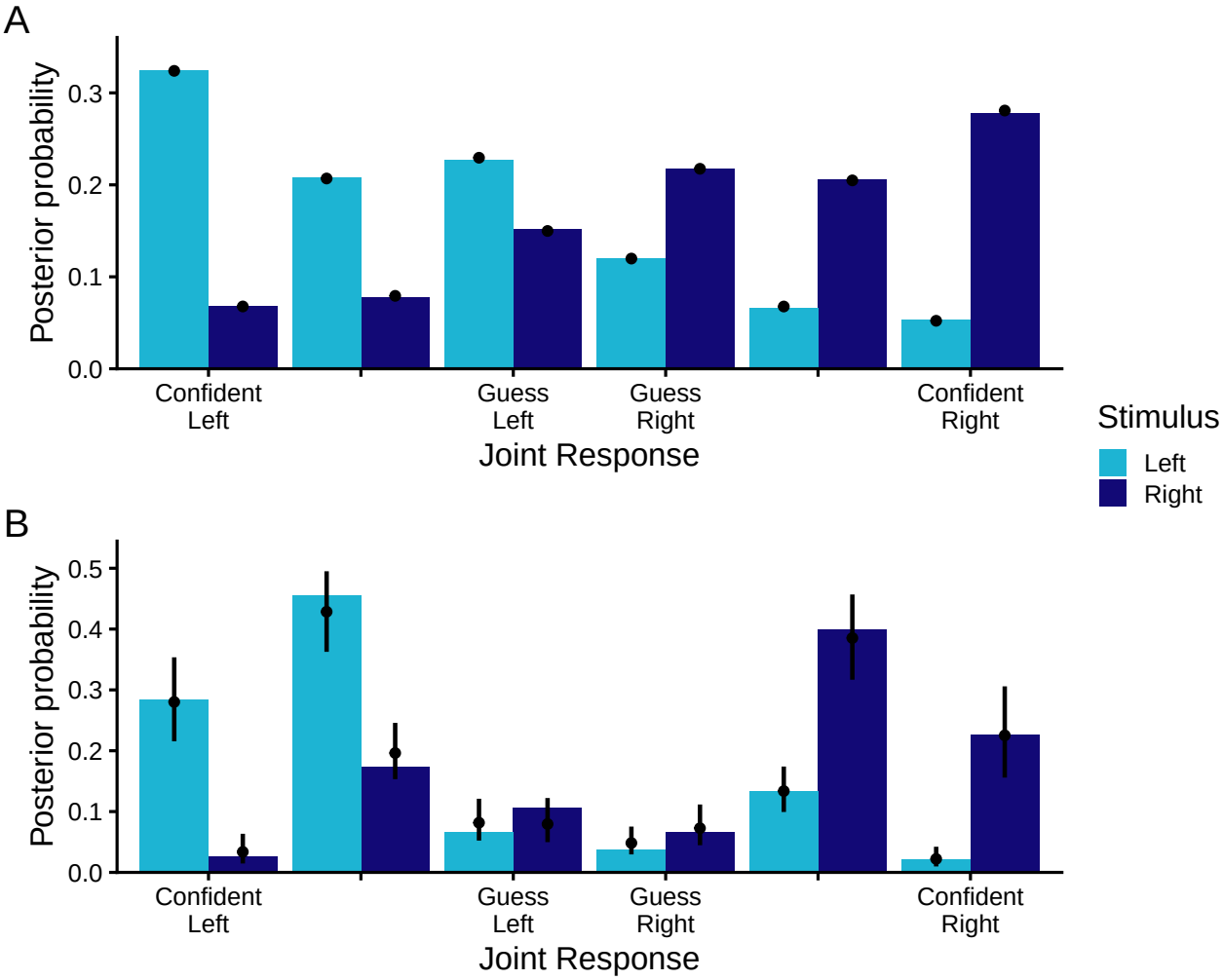
Comparing these probabilities with the empirical probabilities aggregated over participants (Figure 5A), we can see that the model predicts Type I and Type II responses quite well. We can also investigate model predictions for individual participants to get an even more informative picture. Looking at an example participant in Figure 5B, we see that the model predictions are somewhat uncertain but also fit the data well. Note that posterior retrodictive checks such as these are meant only to identify model degeneracies that are relevant for the research question at hand. In cases where the model has clear predictive limitations, then, it is up to the researcher to decide whether those limitations reflect violations of relevant modeling assumptions that should be revised before proceeding, or if instead the limitations reflect details outside of the scope of the current project.

There are countless other kinds of model quantities and visualizations that can contribute to a more complete picture of the empirical adequacy of our model for our research question, including predictions of Type I and Type II probabilities, Type I and Type II Receiver Operating Characteristic curves (ROCs), area under the Type I and Type II ROCs, and mean confidence. Because all of these quantities use a shared interface, checking them involves a similar process. For example, we can also compute Type II ROCs (plotted in Figure 6):

```
draws.roc2 <- roc2_rvars(
  m, newdata,
  re_formula = NA, bounds = TRUE,
  .response = "response_left", by_response = FALSE
)
draws.roc2
```

Figure 5

Posterior predictions of joint Type I/Type II probabilities, i.e., $P(C = c, R = r \mid S = s)$. (A) *Posterior predictions and data averaged across participants.* (B) *Posterior predictions and data for a single example participant*



Note. Bars depict empirical probabilities and points and intervals depict posterior medians and 95% credible intervals.

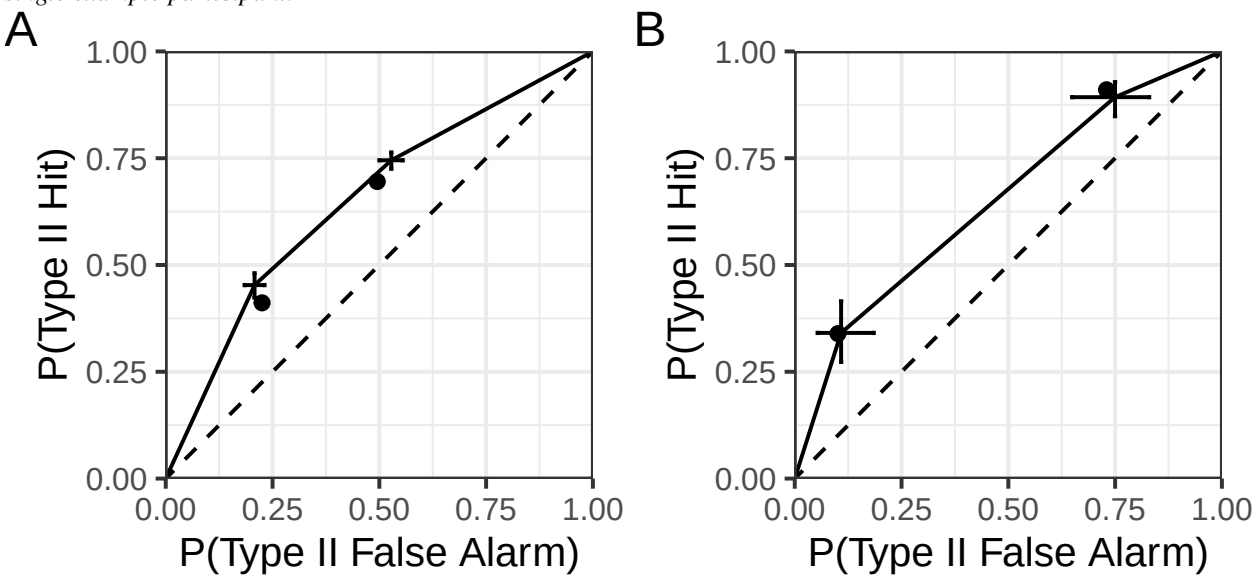
```
# A tibble: 4 x 5
# Groups:   .row, anxious_depression, confidence [4]
   .row anxious_depression confidence         p_hit2         p_fa2
  <int>              <dbl>      <int>     <rvar[1d]>    <rvar[1d]>
1     1                  0          0  1.00 ± 0.0000  1.00 ± 0.000
2     1                  0          1  0.74 ± 0.0098  0.53 ± 0.014
3     1                  0          2  0.45 ± 0.0147  0.21 ± 0.012
4     1                  0          3  0.00 ± 0.0000  0.00 ± 0.000
```

Notably, while each of these implied quantities can help understand the limitations of the model, they are all derived from the joint Type I/Type II response probabilities determined by the signal detection theoretic parameters.

Now that we have verified that our model can describe the overall distribution of participants' responses and confidence ratings, we can inspect the model to answer the research question posed by Rouault et al. (2018): how is anxious-depression related to metacognition? As an initial approach, we can display the model summary:

Figure 6

Posterior predictions of Type II Receiver Operating Characteristic curves (ROCs). (A) *Posterior predictions when all predictors are set to zero compared to data aggregated across participants.* (B) *Posterior predictions and data for a single example participant.*



Note. Bars depict complementary cumulative posterior probabilities and points depict empirical probabilities.

```
summary(m)
```

```
Multilevel Hyperparameters:
~participant (Number of levels: 453)
                                                    Estimate Est.Error
sd(Intercept)                                           0.34      0.02
sd(dprime_Intercept)                                    0.01      0.01
sd(c_Intercept)                                         0.33      0.01
sd(metac2zero1diff_Intercept)                           0.79      0.03
sd(metac2zero2diff_Intercept)                           0.66      0.03
sd(metac2one1diff_Intercept)                            0.74      0.03
sd(metac2one2diff_Intercept)                            0.65      0.03
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)    0.44      0.04
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)     0.92      0.01
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)     0.43      0.04
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)     0.37      0.04
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)     0.95      0.01
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)      0.38      0.04
                                                    l-95% CI u-95% CI Rhat
sd(Intercept)                                           0.31     0.38 1.00
sd(dprime_Intercept)                                    0.00     0.04 1.00
sd(c_Intercept)                                         0.31     0.36 1.00
sd(metac2zero1diff_Intercept)                           0.73     0.85 1.00
sd(metac2zero2diff_Intercept)                           0.61     0.71 1.00
sd(metac2one1diff_Intercept)                            0.69     0.80 1.00
sd(metac2one2diff_Intercept)                            0.60     0.70 1.00
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)    0.35     0.52 1.01
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)     0.90     0.94 1.00
```

```
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)      0.35     0.51 1.00
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)      0.28     0.46 1.00
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)      0.93     0.97 1.00
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)       0.29     0.46 1.00
                                                          Bulk_ESS Tail_ESS
sd(Intercept)                                                 8054    12112
sd(dprime_Intercept)                                          6876     7767
sd(c_Intercept)                                               2872     4878
sd(metac2zero1diff_Intercept)                                 1858     4389
sd(metac2zero2diff_Intercept)                                 2687     5216
sd(metac2one1diff_Intercept)                                  2351     4934
sd(metac2one2diff_Intercept)                                  3047     5630
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)      1984     4056
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)       3411     7650
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)       2515     4728
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)       2168     4487
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)       4547     9154
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)        2757     5403

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI Rhat
```

This summary is divided into two sections: `Multilevel Hyperparameters` and `Regression Coefficients`. The former displays between-participant standard deviations for each model parameter, as well as between-participant correlations among parameters. The latter displays the coefficients for each parameter describing the population. In both sections, `Estimate` is the posterior mean, `Est.Error` is the posterior standard deviation, `l-95% CI` and `u-95% CI` are the 95% posterior quantiles (i.e., credible intervals), and `Rhat`, `Bulk_ESS` and `Tail_ESS` are convergence diagnostics. Under `Multilevel Hyperparameters`, we find that the participant-level standard deviations are quite small for most parameters, though larger for the confidence criteria, indicating significant participant-level variability in metacognitive bias. In particular, the participant-level standard deviation of d' is close to zero, which reflects the fact that the staircase used to control performance across participants in this experiment was quite successful. We also see strong correlations between participant-level estimates of the confidence criteria, which suggests that confidence responding is highly similar regardless of whether participants respond "left" or "right".

Looking at the model intercepts, we see generally suboptimal metacognitive efficiency (with $\log(M) < 0$), good task performance overall (with $d' \approx 1.3$), and a slight bias for the left response option (with $c < 0$). A preliminary inspection of the regression coefficients suggests that there are no effects of anxious-depression on M-ratio, d', or c. In contrast, the distances between confidence criteria appear to increase with anxious-depression, at least for the first confidence level.

We can first visualize the group-level estimate of M-ratio compared to its prior distribution:

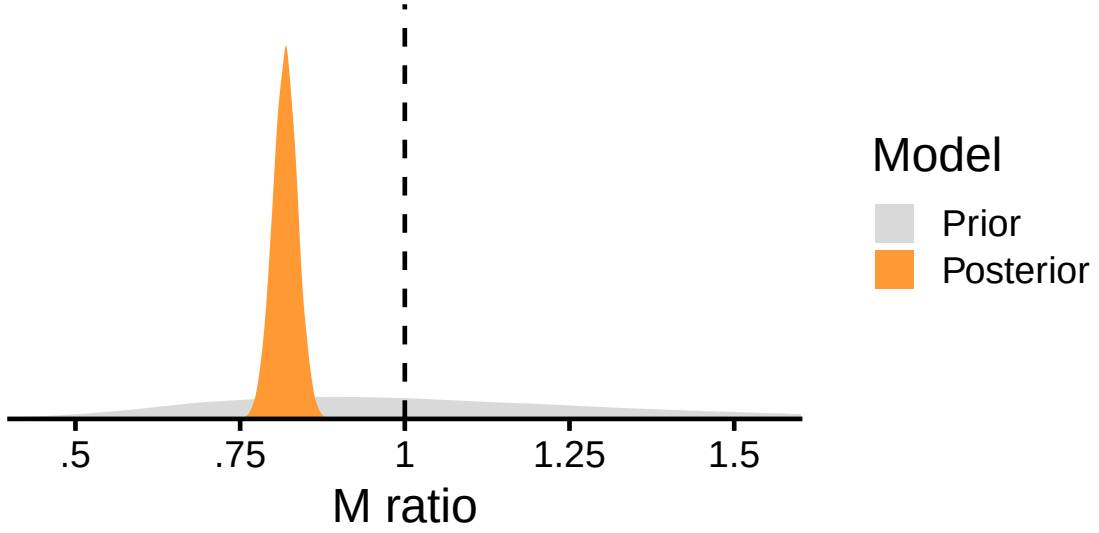
```
draws.mratio <- bind_rows(
  linpred_rvars_metad(m, newdata, re_formula = NA) |>
    mutate(model = "posterior"),
  linpred_rvars_metad(m.prior, newdata, re_formula = NA) |>
    mutate(model = "prior")
)
```

As seen in Figure 7, the posterior distribution of M-ratio is much narrower than our prior, meaning that our data are very informative for estimating this parameter. This visualization also confirms the model summary above indicating that M-ratio is significantly below its optimal value.

Next, in order to interpret the regression coefficients further, we will compute model predictions at varying levels of anxious-depression:

Figure 7

Group-level prior (gray) and posterior (orange) distributions of M-ratio when anxious-depression is at its mean.




```
newdata.ad <- tibble(anxious_depression = seq(-2, 3, by = .25))
draws.linpred <- linpred_rvars_metad(m, newdata.ad, re_formula = NA, pivot_longer = TRUE)
draws.linpred
```

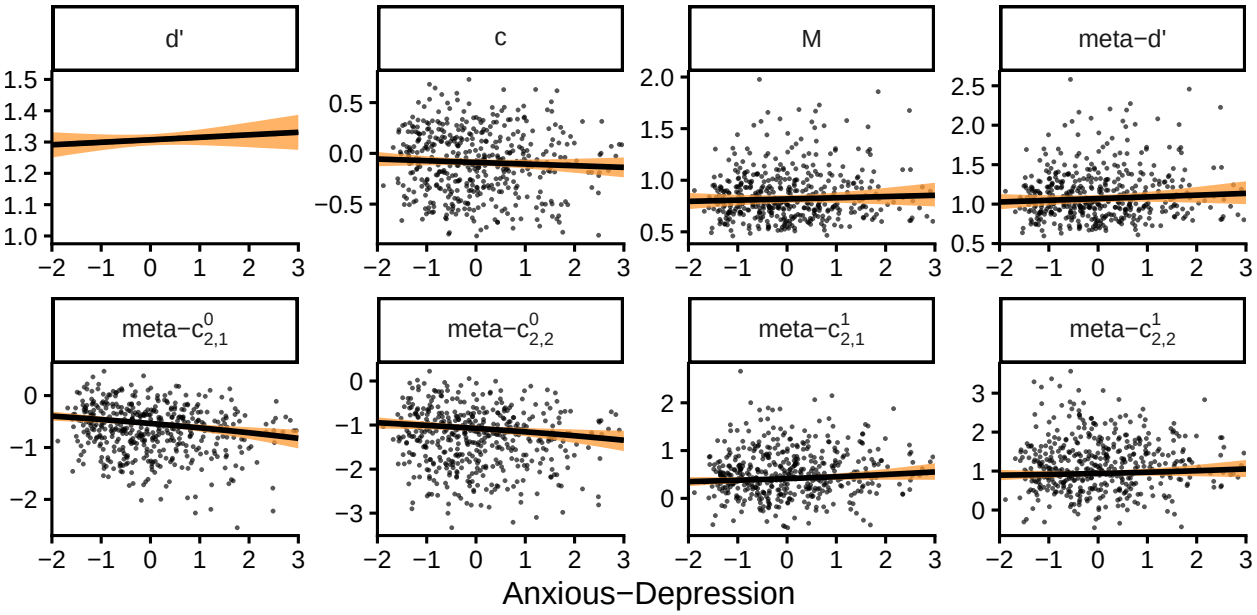
```
# A tibble: 189 x 4
# Groups:   .row, anxious_depression, .variable [189]
   anxious_depression  .row .variable            .value
                <dbl> <int> <chr>             <rvar[1d]>
 1               -2        1 M              0.796 ± 0.040
 2               -2        1 dprime         1.291 ± 0.020
 3               -2        1 c             -0.056 ± 0.036
 4               -2        1 meta_dprime    1.028 ± 0.049
 5               -2        1 meta_c        -0.056 ± 0.036
 6               -2        1 meta_c2_0_1   -0.395 ± 0.045
 7               -2        1 meta_c2_0_2   -0.949 ± 0.065
 8               -2        1 meta_c2_1_1    0.351 ± 0.047
 9               -2        1 meta_c2_1_2    0.896 ± 0.067
10               -1.75     2 M              0.799 ± 0.036
# i 179 more rows
```

Shown in Figure 8, it is clear that the relationships between mental health and M-ratio, d', and c are small if non-existent. In contrast, there appear to be effects on the confidence criteria, however this is complicated because (a) there are two confidence criteria for each Type I response, (b) the confidence criteria for each response are ordered in opposite directions, and (c) effects on the the confidence criteria include potential effects on the Type I criterion, c. To simplify things further, we can compute a corrected measure of metacognitive bias derived from the confidence criteria known as meta-Δ (O'Neill et al., 2026):

```
draws.bias <- metacognitive_bias_rvars(
  m, newdata.ad,
  re_formula = NA, by_response = FALSE
)
draws.bias
```

Figure 8

Model estimates of Type I and Type II parameters as a function of anxious-depression.



Note. Lines indicate posterior medians and ribbons indicate 95% posterior quantiles. Points indicate posterior medians of participant-level estimates.

```
# A tibble: 21 x 3
# Groups:   .row, anxious_depression [21]
    .row anxious_depression metacognitive_bias
   <int>              <dbl>         <rvar[1d]>
 1     1              -2          0.65 ± 0.040
 2     2              -1.75       0.66 ± 0.037
 3     3              -1.5        0.67 ± 0.034
 4     4              -1.25       0.68 ± 0.031
 5     5              -1          0.69 ± 0.028
 6     6              -0.75       0.70 ± 0.025
 7     7              -0.5        0.72 ± 0.023
 8     8              -0.25       0.73 ± 0.022
 9     9               0          0.74 ± 0.022
10    10               0.25       0.76 ± 0.023
# i 11 more rows
```

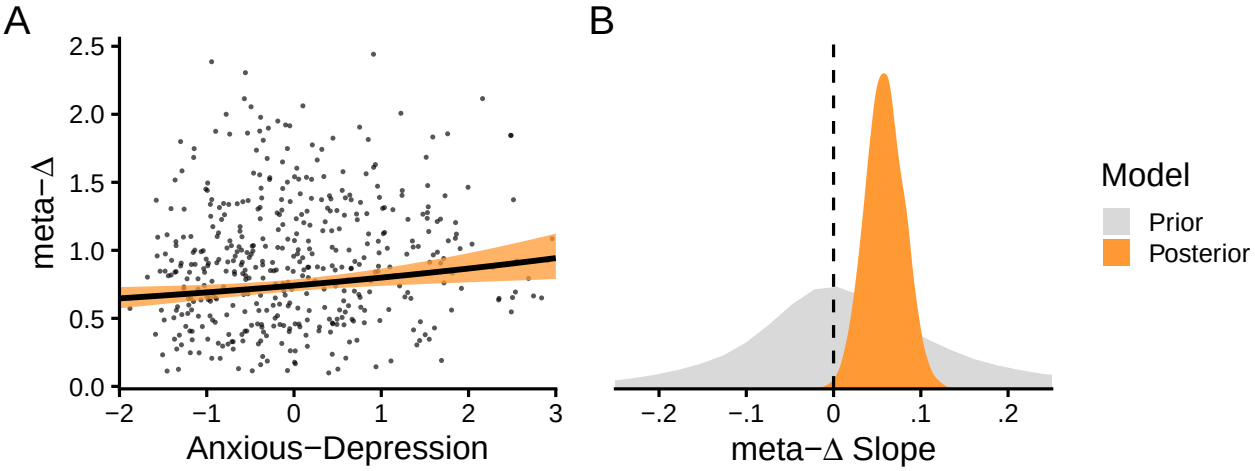
Figure 9 makes clear that increases in anxious-depression are associated with small increases in metacognitive bias (i.e., the amount of evidence required to make a response with confidence at the mid-point).

Significance testing

The final step to answering our research question is to perform significance tests that quantify the amount of evidence for mental-health-related differences in model parameters. Of course, the credible intervals on our previous estimates give us an idea of the probable size of each of these effects. But while credible intervals themselves provide some information about the existence of an effect, they do not provide information about statistical significance (Makowski, Ben-Shachar, Chen, et al., 2019). So, here we will focus on one approach to Bayesian significance testing: Bayes Factors. In general, Bayes Factors represent the degree to which the data have increased the odds of the alternative hypothesis relative to the null hypothesis. As a special case, when using a point null hypothesis (e.g., that anxious-depression has no effect on log M-ratio), it is efficient to compute the Bayes Factor using the Savage-Dickey

Figure 9

Model estimates of metacognitive bias (meta-Δ) as a function of anxious-depression. (A) *Posterior estimates of metacognitive bias at varying levels of anxious-depression.* (B) *Group-level prior (gray) and posterior (orange) distributions of the association between anxious-depression and metacognitive bias.*



Note. Lines indicate posterior medians and ribbons indicate 95% posterior quantiles. Points indicate posterior medians of participant-level estimates.

density ratio, that is, by comparing the prior and posterior density of the null value. This process yields $BF > 1$ when the data provide support for the alternative hypothesis, $BF < 1$ when the data provide support for the null hypothesis, and $BF = 1$ when the data do not inform either hypothesis (for more information, see Makowski, Ben-Shachar, Chen, et al., 2019).

Bayes Factors can be easily computed using the function `describe_posterior` in the `bayestestR` package (Makowski, Ben-Shachar, & Lüdecke, 2019):

```
describe_posterior(m, test = "bf", bf_prior = m.prior, component = "all")
```

```
Summary of Posterior Distribution

Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------------
(Intercept)        |  -0.20 | [-0.25, -0.16] | 7.55e+08 | 1.000 |      13265
anxious_depression |   0.01 | [-0.03,  0.06] |    0.106 | 1.000 |      13443

# Fixed effects (dprime)

Parameter          |   Median |         95% CI |        BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------------
(Intercept)        |     1.31 | [ 1.29,  1.32] | 9.20e+219 | 1.000 |      15029
anxious_depression | 8.01e-03 | [-0.01,  0.03] |     0.053 | 1.000 |      15637

# Fixed effects (c)

Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------------
(Intercept)        |  -0.09 | [-0.12, -0.06] | 1.63e+04 | 1.006 |       3298
anxious_depression |  -0.02 | [-0.05,  0.01] |    0.108 | 1.001 |       3736

# Fixed effects (metac2zero1diff)
```

```
Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------
(Intercept)        |  -0.80 | [-0.88, -0.73] | 3.15e+20 | 1.004 |       1847
anxious_depression |   0.14 | [ 0.07,  0.21] |   228.36 | 1.006 |       1757

# Fixed effects (metac2zero2diff)

Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------
(Intercept)        |  -0.62 | [-0.68, -0.55] | 1.41e+18 | 1.006 |       2657
anxious_depression |  -0.01 | [-0.07,  0.05] |    0.136 | 1.000 |       2993

# Fixed effects (metac2one1diff)

Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------
(Intercept)        |  -0.69 | [-0.76, -0.62] | 2.92e+17 | 1.003 |       1879
anxious_depression |   0.11 | [ 0.04,  0.17] |    17.04 | 1.006 |       2298

# Fixed effects (metac2one2diff)

Parameter          | Median |         95% CI |       BF |  Rhat | ESS (tail)
-------------------------------------------------------------------------
(Intercept)        |  -0.64 | [-0.71, -0.58] | 6.83e+21 | 1.006 |       2717
anxious_depression |  -0.02 | [-0.08,  0.04] |    0.144 | 1.000 |       3380
```

In line with our visualizations above, the Bayes Factors show evidence for null effects of anxious-depression on M-ratio, d', and c, but evidence for mixed effects on the confidence criteria. Specifically, while the confidence criteria for the lowest confidence level are shifted further away with anxious-depression, we see moderate evidence against effects on the highest confidence level. Again, to get a clearer picture, we can analyze meta-Δ instead of the individual confidence criteria. Since meta-Δ is a derived value from the model (and not directly a modeled parameter), we must compute these contrasts manually using the function `compare_levels` from the `tidybayes` package:

```
contr.bias <- draws.bias |>
  filter(anxious_depression %in% c(0, 1)) |>
  compare_levels(metacognitive_bias, by = anxious_depression)
```

Next, we need to compute the same set of contrasts using our prior model (code omitted for brevity). Finally, after merging the two sets of predictions into a single dataframe, we can compute the Bayes Factors using the function `bayestestR::bf_pointnull`:

```
contr.bias |>
  left_join(prior.bias) |>
  mutate(BF = as.numeric(bf_pointnull(metacognitive_bias, prior)))
```

```
# A tibble: 1 x 4
# Groups:   anxious_depression [1]
  anxious_depression metacognitive_bias         prior    BF
  <chr>                      <rvar[1d]>    <rvar[1d]> <dbl>
1 1 - 0                  0.059 ± 0.022  0.027 ± 0.21  8.16
```

This analysis confirms the findings by Rouault et al. (2018), namely that there is moderately strong evidence that overall confidence level (corrected for Type I confounds using the meta-Δ statistic) is decreased with anxious-depression.

Adding complexity

Above, we have somewhat simplified our analysis by looking only at associations with anxious-depression. But Rouault et al. (2018) found two other relevant mental health factors: compulsivity and social withdrawal. Previous implementations would require manual re-coding of the model to incorporate these additional effects. With the `hmetad` package, however, we can simply include these additional factors as predictors in the model formula and re-fit the model:

```
formula <- bf(
  N ~ anxious_depression + compulsivity + social_withdrawal + (1 | participant),
  dprime + c ~ anxious_depression + compulsivity + social_withdrawal + (1 | participant),
  metac2zero1diff + metac2zero2diff +
    metac2one1diff + metac2one2diff ~
    anxious_depression + compulsivity + social_withdrawal + (1 | p | participant)
)

m.prior.full <- fit_metad(
  formula, d,
  .stimulus = "stimulus_left", .response = "response_left",
  prior = prior, sample_prior = "only", cores = 4, init = 0,
  file = "_models/prior_full.rds"
)

m.full <- update(
  m.prior.full,
  file = "_models/fit_full.rds",
  cores = 4, sample_prior = "no", control = list(adapt_delta = 0.95),
)
```

```
Multilevel Hyperparameters:
~participant (Number of levels: 453)
                                                    Estimate Est.Error
sd(Intercept)                                           0.34      0.02
sd(dprime_Intercept)                                    0.01      0.01
sd(c_Intercept)                                         0.33      0.01
sd(metac2zero1diff_Intercept)                           0.77      0.03
sd(metac2zero2diff_Intercept)                           0.65      0.02
sd(metac2one1diff_Intercept)                            0.73      0.03
sd(metac2one2diff_Intercept)                            0.65      0.02
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)    0.42      0.04
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)     0.92      0.01
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)     0.41      0.04
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)     0.36      0.05
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)     0.95      0.01
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)      0.37      0.04
                                                    l-95% CI u-95% CI Rhat
sd(Intercept)                                           0.30     0.38 1.00
sd(dprime_Intercept)                                    0.00     0.04 1.00
sd(c_Intercept)                                         0.31     0.36 1.01
sd(metac2zero1diff_Intercept)                           0.71     0.83 1.01
sd(metac2zero2diff_Intercept)                           0.61     0.70 1.00
sd(metac2one1diff_Intercept)                            0.68     0.79 1.00
sd(metac2one2diff_Intercept)                            0.60     0.69 1.00
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)    0.33     0.50 1.01
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)     0.90     0.94 1.00
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)     0.33     0.49 1.01
```

```
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)      0.27     0.44 1.01
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)      0.93     0.97 1.00
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)       0.28     0.45 1.01
                                                         Bulk_ESS Tail_ESS
sd(Intercept)                                                2275     2981
sd(dprime_Intercept)                                         2472     2450
sd(c_Intercept)                                               932     1610
sd(metac2zero1diff_Intercept)                                 686     1071
sd(metac2zero2diff_Intercept)                                1048     1420
sd(metac2one1diff_Intercept)                                  830     1866
sd(metac2one2diff_Intercept)                                 1089     1437
cor(metac2zero1diff_Intercept,metac2zero2diff_Intercept)      628     1475
cor(metac2zero1diff_Intercept,metac2one1diff_Intercept)      1469     2224
cor(metac2zero2diff_Intercept,metac2one1diff_Intercept)       861     1960
cor(metac2zero1diff_Intercept,metac2one2diff_Intercept)       757     1468
cor(metac2zero2diff_Intercept,metac2one2diff_Intercept)      1402     2410
cor(metac2one1diff_Intercept,metac2one2diff_Intercept)        965     1947

Regression Coefficients:
                              Estimate Est.Error l-95% CI u-95% CI Rhat
Intercept                        -0.20      0.02    -0.25    -0.15 1.00
dprime_Intercept                  1.31      0.01     1.29     1.33 1.00
c_Intercept                      -0.09      0.02    -0.12    -0.06 1.01
metac2zero1diff_Intercept        -0.80      0.04    -0.87    -0.73 1.00
metac2zero2diff_Intercept        -0.62      0.03    -0.68    -0.55 1.00
metac2one1diff_Intercept         -0.69      0.04    -0.76    -0.62 1.00
metac2one2diff_Intercept         -0.64      0.03    -0.70    -0.58 1.00
anxious_depression                0.00      0.03    -0.05     0.05 1.00
compulsivity                     -0.01      0.02    -0.05     0.04 1.00
social_withdrawal                 0.03      0.02    -0.02     0.08 1.00
dprime_anxious_depression         0.00      0.01    -0.02     0.03 1.00
dprime_compulsivity               0.01      0.01    -0.01     0.02 1.00
dprime_social_withdrawal          0.00      0.01    -0.02     0.02 1.00
c_anxious_depression             -0.03      0.02    -0.06     0.00 1.00
```

As we can see from the model summary, the other mental health factors also appear only to affect the confidence criteria. As before, we can visualize group-level estimates of metacognitive bias at varying levels of each factor while holding the others constant:

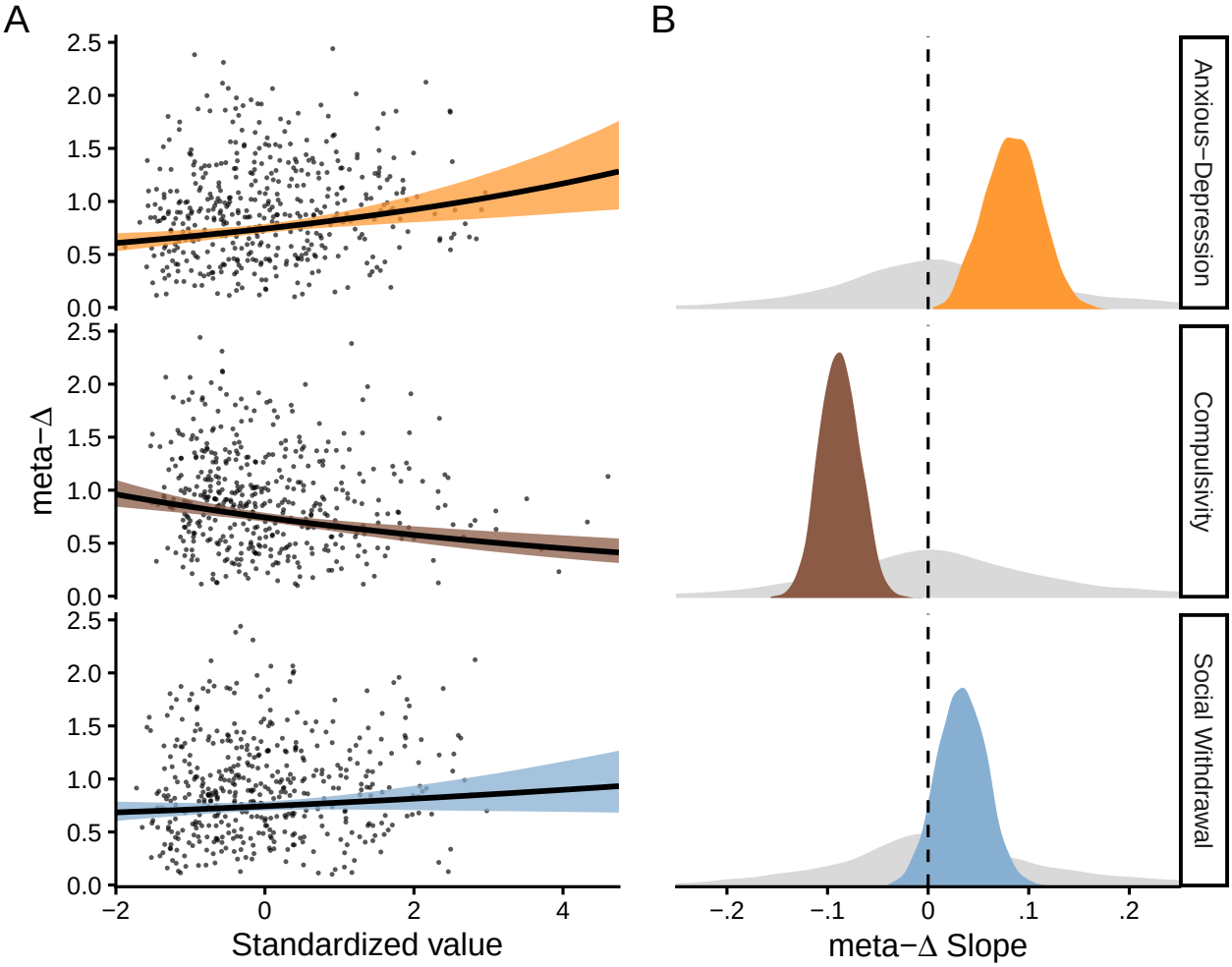
```
mh_grid <- seq(-2, 4.75, by = .25)
newdata.ad <- tibble(
  anxious_depression = mh_grid, compulsivity = 0, social_withdrawal = 0
)
newdata.co <- tibble(
  anxious_depression = 0, compulsivity = mh_grid, social_withdrawal = 0
)
newdata.sw <- tibble(
  anxious_depression = 0, compulsivity = 0, social_withdrawal = mh_grid
)

draws.bias.ad <- metacognitive_bias_rvars(
  m.full, newdata.ad,
  re_formula = NA, by_response = FALSE
)
```

```
draws.bias.co <- metacognitive_bias_rvars(
  m.full, newdata.co,
  re_formula = NA, by_response = FALSE
)
draws.bias.sw <- metacognitive_bias_rvars(
  m.full, newdata.sw,
  re_formula = NA, by_response = FALSE
)
```

Figure 10

Model estimates of metacognitive bias (meta-Δ) as a function of anxious-depression, compulsivity, and social withdrawal. (A) *Posterior estimates of metacognitive bias at varying levels of anxious-depression.* (B) *Group-level prior (gray) and posterior (orange) distributions of the association between anxious-depression and metacognitive bias.*



Note. Lines indicate posterior medians and ribbons indicate 95% posterior quantiles. Points indicate posterior medians of participant-level estimates.

Finally, once again we can compute Bayes Factors for these contrasts:

```
contr.bias |>
  mutate(BF = as.numeric(bf_pointnull(posterior, prior)))
```

```
# A tibble: 3 x 4
```

```
  factor                  posterior         prior      BF
  <chr>                  <rvar[1d]>    <rvar[1d]>   <dbl>
1 Anxious-Depression   0.084 ± 0.027  0.028 ± 0.21  86.0
2 Compulsivity        -0.088 ± 0.020  0.020 ± 0.21 728.
3 Social Withdrawal    0.034 ± 0.024  0.032 ± 0.20   0.617
```

When including all three factors in the model, we now see very strong evidence that confidence levels are decreased with anxious-depression and increased with compulsivity.

We end this section by pointing out that the analysis above is merely an example of the kinds of analyses one can perform with the `hmetad` package: depending on one's research question, experimental design, and preferences for statistical tests and software, one is able to perform a wide range of statistical analyses on the fitted model.

Discussion

Although the field has made significant progress, operationalization, measurement, and modeling are still key obstacles in metacognition research. The meta-d' model addresses this challenge by independently estimating participants' sensitivity to the stimulus when making a response and when rating confidence (Maniscalco & Lau, 2012, 2014). However, despite advances in Bayesian estimation techniques (Fleming, 2017), the practical details of model fitting, visualization, and significance testing have continued to complicate analysis of confidence data.

Here, we show how the `hmetad` package allows users to fit the meta-d' model in a flexible, reproducible, and principled manner. As an extension of the popular `brms` package in R, the `hmetad` package encourages users to set weakly informative priors, validate model fit, and interpret model estimates within a standard Bayesian workflow (Gelman et al., 2026). Besides helping to overcome limitations in the statistical power and reliability of metacognitive efficiency and bias estimates (Fleming, 2017; Rahnev, 2025), this package expands the scope of the model to include trial-level effects, non-linear terms, crossed random effects, correlations among participant-level estimates of Type I and Type II parameters, joint modeling of confidence and other measures, mediation analyses, and many other designs. Although we have not discussed them here, it also enables a broader set of practices in statistical modeling including simulation-based calibration (Modrák et al., 2025; Talts et al., 2018), model comparison (Sivula et al., 2025; Vehtari et al., 2017), and prior sensitivity analysis (Kallioinen et al., 2023, 2026). Overall, the `hmetad` supports analytical practices that address the growing complexity of research questions in metacognitive science.

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Appendix
Log likelihood of the meta-d' model

Computation with raw probabilities often leads to numerical issues like underflow, as well as potentially problematic posterior geometry. To reduce both issues, the probabilistic programming language Stan (which the `hmetad` pacakge is built on) performs computations and evaluates gradients on the logarithmic scale. For the categorical likelihood:

$$\begin{aligned}\ell(\boldsymbol{\theta} \mid S = \boldsymbol{s}, R = \boldsymbol{r}, C = \boldsymbol{c}) &= \log \mathcal{L}(\boldsymbol{\theta} \mid S = \boldsymbol{s}, R = \boldsymbol{r}, C = \boldsymbol{c}) \\ &= \log \prod_{n=1}^{N} P(R = r_n, C = c_n \mid S = s_n) \\ &= \sum_{n=1}^{N} \log P(R = r_n, C = c_n \mid S = s_n)\end{aligned}$$

And for the multinomial likelihood:

$$\begin{aligned}\ell(\boldsymbol{\theta} \mid \boldsymbol{N}) &= \log \mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{N}) \\ &\propto \log \prod_{s,r,c} P(R = r, C = c \mid S = s)^{N_{s,r,c}} \\ &= \sum_{s,r,c} N_{s,r,c} \log P(R = r, C = c \mid S = s)\end{aligned}$$

In both cases, the likelihood reduces to a sum over the log joint probability of a Type I and Type II response given a stimulus. To further increase numerical stability, the `hmetad` package also computes these probabilities on the logarithmic scale. Where $F_{\mathcal{D}}$ is the cumulative distribution function of $\mathcal{D}$ and $\bar{F}_{\mathcal{D}}$ is its complementary cumulative distribution function:

$$
\begin{aligned}
&\log P(R = r, C = c \mid S = s) \\
&\quad = \log P(R = r \mid S = s) + \log P(C = c \mid S = s, R = r) \\
&\quad = \begin{cases}
\log\left[F_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[F_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c) - F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^0_{2,1}\right)\right] & \text{if } r = 0, c = 1 \\
\quad - \log\left[F_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \\
\log\left[F_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^0_{2,k}\right) - F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^0_{2,k+1}\right)\right] & \text{if } r = 0, 1 \le c \le K \\
\quad - \log\left[F_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \\
\log\left[F_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^0_{2,K}\right)\right] & \text{if } r = 0, c = K \\
\quad - \log\left[F_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \\
\log\left[\bar{F}_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^1_{2,1}\right) - F_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \text{if } r = 1, c = 1 \\
\quad - \log\left[\bar{F}_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \\
\log\left[\bar{F}_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^1_{2,k+1}\right) - F_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^1_{2,k}\right)\right] & \text{if } r = 1, 1 \le c \le K \\
\quad - \log\left[\bar{F}_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] & \\
\log\left[\bar{F}_{\mathcal{D}_s(d')}(c)\right] & \\
\quad + \log\left[\bar{F}_{\mathcal{D}_s(\text{meta-}d')}\left(\text{meta-}c^1_{2,K}\right)\right] & \text{if } r = 1, c = K \\
\quad - \log\left[\bar{F}_{\mathcal{D}_s(\text{meta-}d')}(\text{meta-}c)\right] &
\end{cases}
\end{aligned}
$$

Here, each term is the logarithm of either the cumulative distribution function, the complementary cumulative distribution function, or the difference between cumulative distribution functions of $\mathcal{D}$. Stan has built-in functions for the logarithm of the (complementary) cumulative distribution function of most distributions (e.g., the standard normal distribution). So, all that is needed is to compute the difference on the log scale. The `hmetad` function accomplishes this through the Stan function $\texttt{log_diff_exp}(x, y) = \log(e^x - e^y)$, in particular:

$$\log\left[F_{\mathcal{D}}(c_1) - F_{\mathcal{D}}(c_2)\right] = \texttt{log_diff_exp}(\log F_{\mathcal{D}}(c_1), \log F_{\mathcal{D}}(c_2))$$

Because of this formulation, the `hmetad` package supports any unbounded Thurstonian distribution (i.e., any distribution with a location parameter and fixed scale) given a definition of its log (complementary) cumulative distribution function in both R and Stan.